



Modeling double bounded data based on correlated gamma random variables

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Abstract

Many types of bounded data defined on the unit interval arise naturally as ratios of the form $X/(X + Y)$. In the existing literature, the main statistical models proposed for this type of bounded data are typically based on the assumption that the random variables X and Y are independent. However, this assumption is often unrealistic in practical applications, where X and Y tend to be correlated due to shared underlying mechanisms or common sources of variability. In this paper, we overcome such limitations and propose a model in which the marginal distributions of the two components are linked by a copula, leading to a more flexible and realistic representation of unit-interval data. In particular, in the proposed model, X and Y are dependent gamma random variables whose joint distribution is specified via Morgenstern's bivariate distribution, allowing for positive and negative correlations between the components. The mathematical properties and practical applications are rigorously investigated. The resulting distribution exhibits a wide range of shapes, accommodating different degrees of skewness and, for some parameter configurations, more complex density structures. A Monte Carlo simulation study is carried out that shows the good performance of the maximum likelihood estimator in several scenarios of parameter choices. The potential and limitations of efficient likelihood-based computations are also discussed. We evaluate the effectiveness of the new model and its estimates in modeling real-world datasets related to economics.

Keywords Extended beta distribution · Morgenstern's bivariate distribution · Monte Carlo simulation · R software

Mathematics Subject Classification Primary 60E05 · Secondary 62Exx, 62Fxx

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1 Introduction

Several natural indicators are measured as indicators, percentages, proportions, ratios, and rates that are bounded on a certain interval, usually in the bounded interval $(0, 1)$. The need for modeling and analyzing bounded data occurs in many fields of real life such as the Human Development Index, COVID-19 recovery rates, and percentages, among others. Certainly, the two-parameter beta distribution is the most common distribution used in the literature to describe data in the unit interval, especially because of its flexibility.

Malik (1967) and Ahuja (1969) both showed that if X and Y are independent random variables following gamma distributions with shape parameters $\alpha > 0$ and $\beta > 0$, and the same scale parameter σ , i.e., $X \sim \text{Gamma}(\alpha, \sigma)$ and $Y \sim \text{Gamma}(\beta, \sigma)$, then

$$Z \stackrel{d}{=} \frac{X}{X+Y} \sim \text{Beta}(\alpha, \beta), \quad (1)$$

namely, Z is beta distributed with shape parameters $\alpha, \beta > 0$ and $\stackrel{d}{=}$ stands for equality in distribution. It should be noted that stochastic representations are important since they may justify some models arising naturally in real situations. Furthermore, models generated by (1) can produce distributions with different shapes, such as symmetry, bimodality, and others. This allows estimation procedures to better capture the inherent structures in the data. However, the independence assumption seems to be not suitable in real situations, where we would expect a relationship between these variables. The case of positively correlated gamma distributed components has earlier been studied by Nadarajah and Kotz (2006) based on the Kibble bivariate gamma distribution for $(X, Y)^\top$ (Kibble 1941). However, negatively correlated measurements are encountered in numerous applications, for example in COVID-19 recovery rates, where X and Y are two random variables representing the number of confirmed COVID-19-related deaths and COVID-19 cases without death result, and the independence assumption seems to be not suitable in this case.

In this context, considering distributed components negatively and positively correlated, and satisfying the stochastic representation in (1), Vila et al. (2024b) proposed a class of unit-log-symmetric models to analyzing an internet access data. Vila et al. (2025) introduced a new distribution in the unit interval where X and Y are two correlated Birnbaum-Saunders random variables. In the same way, Vila and Quintino (2026) considered a novel unit-asymmetric distribution, where X and Y are two correlated Fréchet random variables.

In this work, following the same stochastic formulation as the beta distribution, we introduce a new class of unit models that arises from a ratio (1) of two correlated gamma random variables. We consider $(X, Y)^\top$ distributed as Morgenstern's (which is also known in the literature as the Farlie-Gumbel-Morgenstern (FGM)) bivariate distribution with gamma margins (cf. Lai 1978). The new model extends the well-known Beta distribution with the addition of one parameter. However, this extra parameter enables the model to exhibit U- or J-shapes, symmetry, as well as more complex behaviors for certain parameter configurations. Moreover, information criteria on real-world dataset applications show that better fits are obtained with the new model, even with this extra parameter. Maximum likelihood-based estimation procedures are discussed for the new distribution. A Monte Carlo simulation study is carried out, as well as applications to datasets well-consolidated in the literature, showing the versatility of the new model. It is well known that the Morgenstern copula is suitable for modeling bivariate data with weak dependence. However, it allows for

analytically tractable derivations of the properties of the corresponding unit model. Moreover, our applications show that the proposed unit model can provide a good fit even when the original data exhibit strong correlation.

Our results rely on special functions and their properties, which let us explicitly derive the probability density function (PDF), the cumulative distribution function (CDF), and key characteristics of the new probabilistic model. Special functions are widely studied because they encompass many classes of functions and provide useful identities for integration, differentiation, and other operations. Some well-known examples are the Gauss hypergeometric ${}_2F_1$ function, Meijer's G -function, Fox's H -function, modified Bessel K_ν -function of the 3rd kind (see Definitions 1.1, 1.5 and 1.6 and 1.11 in Mathai et al. 2010), and the double hypergeometric Appel functions F_1, F_2, F_3 and F_4 (Appell 1925).

The remainder of this paper is organized as follows: Sect. 2 introduces the new model and plots of its shape for different parameter choices. Section 3 deals with the derivation of the new unit distribution and its mathematical properties, such as symmetry, stochastic representation, characterization as a ratio and moments. In Sect. 4, we propose the estimation of the parameters. In Sect. 5, we discuss Monte Carlo simulations to evaluate the proposed estimator, its computational challenges, and alternative frameworks. In Sect. 6, we address modeling real-world scenarios involving an income consumption dataset. The final section presents our conclusions, while Appendix A, B and C present, respectively, some additional results on the Appel F_2 function, the competing models and some numerical verifications of the CDF implementation.

2 The extended beta distribution

We say that an absolutely continuous random variable Z , with support $(0, 1)$, follows an extended beta (EB) distribution with parameter vector $\theta = (\alpha, \beta, \rho)^\top$, $\alpha > 0, \beta > 0$, and $-1 \leq \rho \leq 1$, denoted by $Z \sim \text{EB}(\theta)$, if its PDF is given by (for $0 < z < 1$)

$$\begin{aligned} f_Z(z; \theta) = & (1 + \rho) \frac{z^{\alpha-1}(1-z)^{\beta-1}}{\text{B}(\alpha, \beta)} \\ & - \rho \frac{2\Gamma(2\alpha + \beta)}{\alpha\Gamma^2(\alpha)\Gamma(\beta)} \frac{z^{2\alpha-1}(1-z)^{\beta-1}}{(1+z)^{2\alpha+\beta}} {}_2F_1\left(1, 2\alpha + \beta; 1 + \alpha; \frac{z}{1+z}\right) \\ & - \rho \frac{2\Gamma(\alpha + 2\beta)}{\beta\Gamma(\alpha)\Gamma^2(\beta)} \frac{z^{\alpha-1}(1-z)^{2\beta-1}}{(2-z)^{\alpha+2\beta}} {}_2F_1\left(1, \alpha + 2\beta; 1 + \beta; \frac{1-z}{2-z}\right) \\ & + \rho \frac{4^{1-\alpha-\beta}\Gamma(2\alpha + 2\beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma^2(\beta)} z^{\alpha-1}(1-z)^{2\beta-1} F_2\left(2\alpha + 2\beta, 1, 1; \alpha + 1, \beta + 1; \frac{z}{2}, \frac{1-z}{2}\right), \end{aligned} \quad (2)$$

where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} \exp(-t) dt$ is the complete gamma function, $\text{B}(\alpha_1, \alpha_2) = \Gamma(\alpha_1)$

$\Gamma(\alpha_2)/\Gamma(\alpha_1 + \alpha_2)$ is the complete beta function,

$${}_2F_1(a, b; c; x) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{x^n}{n!}$$

is the hypergeometric function,

$$F_2(a, b_1, b_2; c_1, c_2; x, y) = \sum_{m,n=0}^{\infty} \frac{(a)_{m+n} (b_1)_m (b_2)_n}{(c_1)_m (c_2)_n} \frac{x^m}{m!} \frac{y^n}{n!} \quad (3)$$

denotes the Appell F2 double hypergeometric function, and $(q)_n = \Gamma(q+n)/\Gamma(q)$ is the Pochhammer symbol. Conditions for the convergence of the series and several properties of these special functions are given in Mathai et al. (2010) and Ananthanarayan et al. (2023).

Remark 2.1 Note that when $\rho = 0$ the EB model in (2) becomes the classic beta distribution. Therefore, the EB model can be interpreted as a new extended beta distribution.

Figure 1 shows the PDF $f_Z(z; \theta)$ behavior for some parameter choices. Note that we can have a symmetric distribution centered around 0.5 (if $\alpha = \beta$, see Fig. 2). It is also possible to obtain asymmetric or bimodal (Fig. 3) models, depending on the choice of parameters.

In Figs. 1, 2 and 3 we note that:

- for $\alpha > \beta$, the shapes are concentrated on the right side of the interval $[0, 1]$;
- for $\alpha < \beta$, the concentration is in the left side;
- when $\rho \approx -1$, bimodality can be observed. However, the two modes are not completely separated. After the first mode, the frequency does not drop near zero before rising again toward the second mode.

Remark 2.2 Usual techniques for establishing unimodality or bimodality rely on analyzing the sign of the PDF's derivative and its roots (see, e.g., Theorem 2.1 in Vila et al. 2024a). However, the PDF (2) involves the special functions ${}_2F_1$ and F_2 , which can make root analysis either impossible or extremely difficult from an analytical approach.

As closed-form conditions are unattainable, a more systematic numerical exploration of the bimodality region is introduced in Algorithm 1. The algorithm can be run over a pre-specified parameter grid $\Theta = \{\theta^{(1)}, \dots, \theta^{(k)}\}$. Furthermore, an additional step can be incorporated

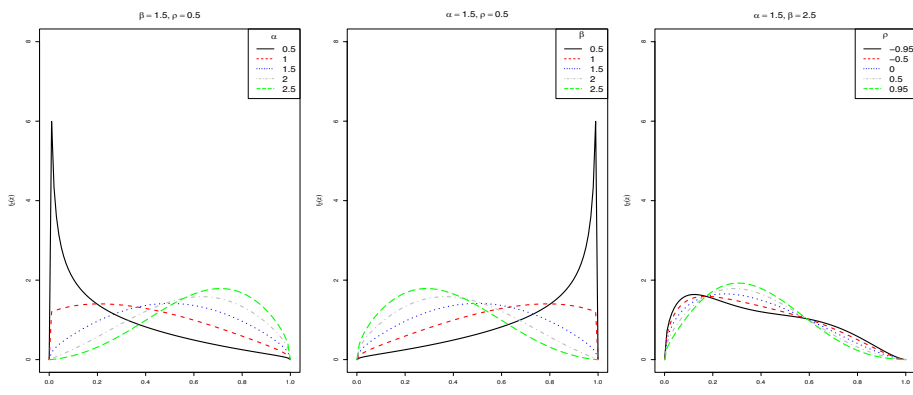


Fig. 1 Plot of the PDF f_Z with varying parameters α (left), β (middle) and ρ (right)

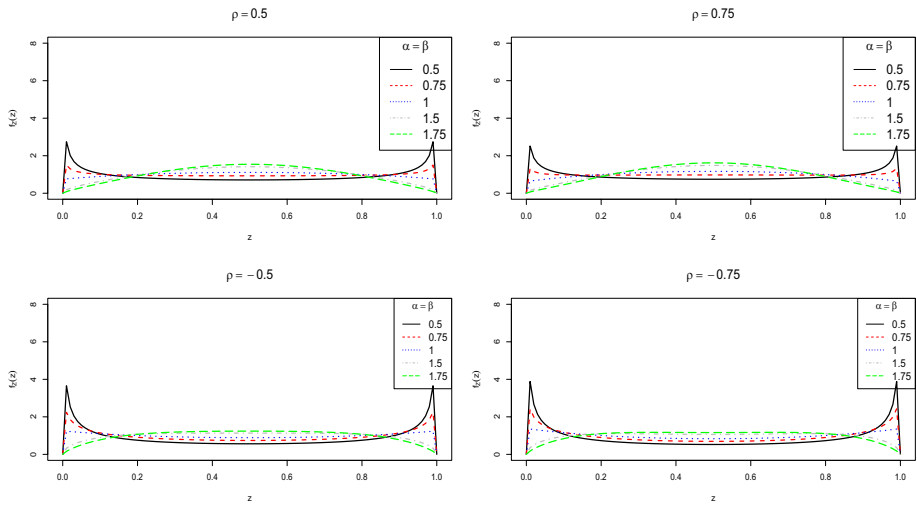


Fig. 2 Plot of the PDF f_Z with varying parameters $\alpha = \beta$ for $\rho \in \{-0.5, -0.75, 0.5, 0.75\}$

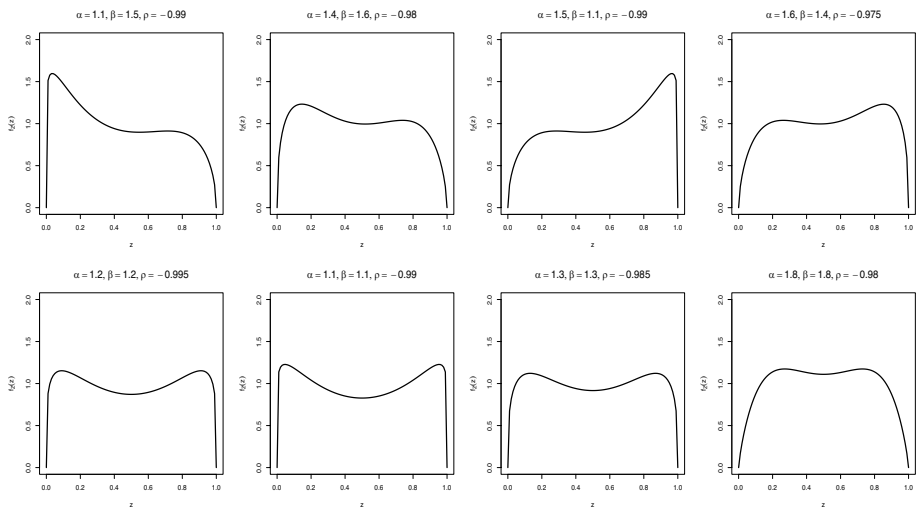


Fig. 3 Plot of the PDF f_Z with varying parameters α , β and ρ , illustrating bimodality

to evaluate the depth of the valley between the two detected peaks. The parameters shown in Fig. 3 were selected after several rounds of testing and refinement of Algorithm 1.

Input: a vector $\theta \in (0, \infty) \times (0, \infty) \times [-1, 1]$;
Output: points of possible bimodality (if they exist).

- 1: Fix the values $z_1, z_2, \dots, z_n \in [0, 1]$;
- 2: Define the lists

$$P \leftarrow \text{NULL}; V \leftarrow \text{NULL};$$
- 3: Compute $\Delta_i f = f(z_{i+1}; \theta) - f(z_i; \theta)$, for $i = 1, \dots, n-1$;
- 4: **for** $i = 2, \dots, n-1$ **do**
- 5: **if** $\Delta_{i-1} > 0$ & $\Delta_i < 0$ **then**
- 6: $P \leftarrow z_i$;
- 7: **end if**
- 8: **if** $\Delta_{i-1} < 0$ & $\Delta_i > 0$ **then**
- 9: $V \leftarrow z_i$;
- 10: **end if**
- 11: **end for**
- 12: **if** $\#P \geq 2$ **then**
- 13: Compute

$$p_1 = \arg \max_{z_i \in P} f(z_i; \theta), \quad p_2 = \arg \max_{z_i \in P; \quad z_i \neq p_1} f(z_i; \theta) \quad \text{and} \quad v_* = \arg \min_{p_2 < z_i < p_1} f(z_i; \theta);$$
- 14: **Return** p_1, p_2 , and v_* .
- 15: **else**
- 16: **Return** “unimodal”.
- 17: **end if**

Algorithm 1 Searching for bimodality

3 Some basic properties

In this section we establish some basic mathematical properties of the extended beta distribution.

3.1 Symmetry

Let $Z \sim \text{EB}(\theta)$. A simple observation shows that (for $0 < w < 1/2$)

$$\begin{aligned}
 f_Z(z; \theta) \Big|_{z=\frac{1}{2}-w} &= (1+\rho) \frac{\left(\frac{1}{2}-w\right)^{\alpha-1} \left(\frac{1}{2}+w\right)^{\beta-1}}{\text{B}(\alpha, \beta)} \\
 &- \rho \frac{2\Gamma(2\alpha+\beta)}{\alpha\Gamma^2(\alpha)\Gamma(\beta)} \frac{\left(\frac{1}{2}-w\right)^{2\alpha-1} \left(\frac{1}{2}+w\right)^{\beta-1}}{\left(\frac{3}{2}-w\right)^{2\alpha+\beta}} {}_2F_1\left(1, 2\alpha+\beta; 1+\alpha; \frac{1-2w}{3-2w}\right) \\
 &- \rho \frac{2\Gamma(\alpha+2\beta)}{\beta\Gamma(\alpha)\Gamma^2(\beta)} \frac{\left(\frac{1}{2}-w\right)^{\alpha-1} \left(\frac{1}{2}+w\right)^{2\beta-1}}{\left(\frac{3}{2}+w\right)^{\alpha+2\beta}} {}_2F_1\left(1, \alpha+2\beta; 1+\beta; \frac{1+2w}{3+2w}\right) \\
 &+ \rho \frac{4^{1-\alpha-\beta}\Gamma(2\alpha+2\beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma^2(\beta)} \left(\frac{1}{2}-w\right)^{2\alpha-1} \left(\frac{1}{2}+w\right)^{2\beta-1} F_2\left(2\alpha+2\beta, 1, 1; \alpha+1, \beta+1; \frac{1-2w}{4}, \frac{1+2w}{4}\right)
 \end{aligned} \tag{4}$$

and

$$\begin{aligned}
 f_Z(z; \theta) \Big|_{z=\frac{1}{2}+w} &= (1+\rho) \frac{(\frac{1}{2}+w)^{\alpha-1}(\frac{1}{2}-w)^{\beta-1}}{B(\alpha, \beta)} \\
 &- \rho \frac{2\Gamma(2\alpha+\beta)}{\alpha\Gamma^2(\alpha)\Gamma(\beta)} \frac{(\frac{1}{2}+w)^{2\alpha-1}(\frac{1}{2}-w)^{\beta-1}}{(\frac{3}{2}+w)^{2\alpha+\beta}} {}_2F_1\left(1, 2\alpha+\beta; 1+\alpha; \frac{1+2w}{3+2w}\right) \\
 &- \rho \frac{2\Gamma(\alpha+2\beta)}{\beta\Gamma(\alpha)\Gamma^2(\beta)} \frac{(\frac{1}{2}+w)^{\alpha-1}(\frac{1}{2}-w)^{2\beta-1}}{(\frac{3}{2}-w)^{\alpha+2\beta}} {}_2F_1\left(1, \alpha+2\beta; 1+\beta; \frac{1-2w}{3-2w}\right) \\
 &+ \rho \frac{4^{1-\alpha-\beta}\Gamma(2\alpha+2\beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma^2(\beta)} \left(\frac{1}{2}+w\right)^{2\alpha-1} \left(\frac{1}{2}-w\right)^{2\beta-1} F_2\left(2\alpha+2\beta, 1, 1; \alpha+1, \beta+1; \frac{1+2w}{4}, \frac{1-2w}{4}\right).
 \end{aligned} \tag{5}$$

As the the Appell F_2 double hypergeometric function is symmetric, that is, $F_2(a, b_1, b_2; c_1, c_2; x, y) = F_2(a, b_1, b_2; c_1, c_2; y, x)$, by taking $\alpha = \beta$ in the identities (4) and (5), we have (for $0 < w < 1/2$)

$$f_Z(z; \theta) \Big|_{z=\frac{1}{2}-w} = f_Z(z; \theta) \Big|_{z=\frac{1}{2}+w}.$$

That is, the EB PDF in (2) is symmetric around $z_0 = 1/2$ provided $\alpha = \beta$ (see Fig. 2). Moreover, in this case, the median and the mean of a EB distribution both occur at $z_0 = 1/2$.

3.2 EB model arising as a ratio of correlated random variables

In this subsection, we establish a fundamental property of the EB model, in which the random variable Z in (2) is represented as a ratio of the form

$$Z \stackrel{d}{=} \frac{X}{X+Y}, \tag{6}$$

where $\stackrel{d}{=}$ means equality in distribution and the random vector $(X, Y)^\top$ follows a Morgenstern's bivariate distribution (with gamma margins) (Lai 1978).

Define the bivariate function $C : [0, 1] \times [0, 1] \rightarrow [0, 1]$ as

$$C(u_1, u_2) = u_1 u_2 [1 + \rho(1 - u_1)(1 - u_2)], \quad (u_1, u_2) \in [0, 1] \times [0, 1] \text{ and } -1 \leq \rho \leq 1. \tag{7}$$

Although the copula $C(\cdot, \cdot)$ defined above has limited dependence flexibility, it was adopted mainly due to its analytical tractability. In order to provide richer dependence structures, we have included in the real-data analysis (Sect. 6) some additional flexible extensions of the copula $C(\cdot, \cdot)$.

Following Lai (1978), a random vector $(X, Y)^\top$ has a Morgenstern's bivariate distribution if its cumulative distribution function (CDF) is of the form

$$F_{X,Y}(x, y) = C(F_X(x), F_Y(y)) = F_X(x)F_Y(y)[1 + \rho\{1 - F_X(x)\}\{1 - F_Y(y)\}], \quad -1 \leq \rho \leq 1, \tag{8}$$

having $F_X(x)$ and $F_Y(y)$ as marginal distribution functions. It is clear that the corresponding joint PDF of $(X, Y)^\top$ is given by

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)[1 + \rho\{2F_X(x) - 1\}\{2F_Y(y) - 1\}]. \tag{9}$$

Note that when $\rho = 0$, X and Y are independent.

In what follows, we verify that the random variable Z in (2) satisfies the stochastic representation (6), where $(X, Y)^\top$ has Morgenstern's bivariate distribution (8) and (9), and X and Y have gamma distributions with the same rate, more precisely, $X \sim \text{Gamma}(\alpha, \theta)$ and $Y \sim \text{Gamma}(\beta, \theta)$, $\alpha > 0$, $\beta > 0$, $\theta > 0$.

By fact, using the Jacobian method, we have (for $s = 1/z - 1$ and $0 < z < 1$)

$$f_{\frac{X}{X+Y}}(z) = (s+1)^2 \int_0^\infty x f_{X,Y}(x, sx) dx.$$

Using (9) in the above identity, we have the following

$$\begin{aligned} f_{\frac{X}{X+Y}}(z) &= (1+\rho)(s+1)^2 \int_0^\infty x f_X(x) f_Y(sx) dx - 2\rho(s+1)^2 \int_0^\infty [x f_X(x) f_Y(sx)] F_X(x) dx \\ &\quad - 2\rho(s+1)^2 \int_0^\infty [x f_X(x) f_Y(sx)] F_Y(sx) dx + 4\rho(s+1)^2 \int_0^\infty [x f_X(x) f_Y(sx)] F_X(x) F_Y(sx) dx \\ &\equiv (1+\rho)K_1(z) - \rho K_2(z) - \rho K_3(z) + \rho K_4(z), \end{aligned} \quad (10)$$

where

$$\begin{aligned} K_1(z) &\equiv (s+1)^2 \int_0^\infty x f_X(x) f_Y(sx) dx, \\ K_2(z) &\equiv 2(s+1)^2 \int_0^\infty [x f_X(x) f_Y(sx)] F_X(x) dx, \\ K_3(z) &\equiv 2(s+1)^2 \int_0^\infty [x f_X(x) f_Y(sx)] F_Y(sx) dx, \\ K_4(z) &\equiv 4(s+1)^2 \int_0^\infty [x f_X(x) f_Y(sx)] F_X(x) F_Y(sx) dx. \end{aligned} \quad (11)$$

Since $X \sim \text{Gamma}(\alpha, \theta)$, $Y \sim \text{Gamma}(\beta, \theta)$ and

$$x f_X(x; \alpha, \theta) f_Y(sx; \beta, \theta) = \frac{\theta^{\alpha+\beta} s^{\beta-1}}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha+\beta-1} \exp[-(1+s)\theta x],$$

we have

$$\begin{aligned} K_1(z) &= (s+1)^2 \frac{\theta^{\alpha+\beta} s^{\beta-1}}{\Gamma(\alpha)\Gamma(\beta)} \int_0^\infty x^{\alpha+\beta-1} \exp[-(1+s)\theta x] dx, \\ K_2(z) &= 2(s+1)^2 \frac{\theta^{\alpha+\beta} s^{\beta-1}}{\Gamma^2(\alpha)\Gamma(\beta)} \int_0^\infty x^{\alpha+\beta-1} \exp[-(1+s)\theta x] \gamma(\alpha, \theta x) dx, \\ K_3(z) &= 2(s+1)^2 \frac{\theta^{\alpha+\beta} s^{\beta-1}}{\Gamma(\alpha)\Gamma^2(\beta)} \int_0^\infty x^{\alpha+\beta-1} \exp[-(1+s)\theta x] \gamma(\beta, \theta sx) dx, \\ K_4(z) &= 4(s+1)^2 \frac{\theta^{\alpha+\beta} s^{\beta-1}}{\Gamma^2(\alpha)\Gamma^2(\beta)} \int_0^\infty x^{\alpha+\beta-1} \exp[-(1+s)\theta x] \gamma(\alpha, \theta x) \gamma(\beta, \theta sx) dx. \end{aligned}$$

It is clear that

$$K_1(z) = \frac{z^{\alpha-1}(1-z)^{\beta-1}}{B(\alpha, \beta)}. \quad (12)$$

By using the identity of Proposition A.2:

$$\int_0^\infty x^{a-1} \exp(-sx) \gamma(b, \theta x) dx = \frac{\theta^b \Gamma(a+b)}{b(s+\theta)^{a+b}} {}_2F_1\left(a+b, 1; b+1; \frac{\theta}{s+\theta}\right), \quad (13)$$

we have

$$K_2(z) = \frac{2\Gamma(2\alpha+\beta)}{\alpha\Gamma^2(\alpha)\Gamma^2(\beta)} \frac{z^{2\alpha-1}(1-z)^{\beta-1}}{(1+z)^{2\alpha+\beta}} {}_2F_1\left(1, 2\alpha+\beta; 1+\alpha; \frac{z}{1+z}\right) \quad (14)$$

and

$$K_3(z) = \frac{2\Gamma(\alpha+2\beta)}{\beta\Gamma(\alpha)\Gamma^2(\beta)} \frac{z^{\alpha-1}(1-z)^{2\beta-1}}{(2-z)^{\alpha+2\beta}} {}_2F_1\left(1, \alpha+2\beta; 1+\beta; \frac{1-z}{2-z}\right). \quad (15)$$

Furthermore, by using the identity of Proposition A.1:

$$\begin{aligned} & \int_0^\infty x^{a-1} \exp(-sx) \gamma(b, \theta x) \gamma(c, \xi x) dx \\ &= \frac{\theta^b \xi^c \Gamma(a+b+c)}{bc(s+\theta+\xi)^{a+b+c}} F_2\left(a+b+c, 1, 1; b+1, c+1; \frac{\theta}{s+\theta+\xi}, \frac{\xi}{s+\theta+\xi}\right), \end{aligned} \quad (16)$$

we obtain

$$K_4(z) = \frac{4^{1-\alpha-\beta} \Gamma(2\alpha+2\beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma^2(\beta)} z^{2\alpha-1}(1-z)^{2\beta-1} F_2\left(2\alpha+2\beta, 1, 1; \alpha+1, \beta+1; \frac{z}{2}, \frac{1-z}{2}\right). \quad (17)$$

By replacing (12), (14), (15) and (17) in (10), we get (for $0 < z < 1$)

$$\begin{aligned} f_{\frac{x}{x+y}}(z) &= (1+\rho) \frac{z^{\alpha-1}(1-z)^{\beta-1}}{B(\alpha, \beta)} \\ &\quad - \rho \frac{2\Gamma(2\alpha+\beta)}{\alpha\Gamma^2(\alpha)\Gamma^2(\beta)} \frac{z^{2\alpha-1}(1-z)^{\beta-1}}{(1+z)^{2\alpha+\beta}} {}_2F_1\left(1, 2\alpha+\beta; 1+\alpha; \frac{z}{1+z}\right) \\ &\quad - \rho \frac{2\Gamma(\alpha+2\beta)}{\beta\Gamma(\alpha)\Gamma^2(\beta)} \frac{z^{\alpha-1}(1-z)^{2\beta-1}}{(2-z)^{\alpha+2\beta}} {}_2F_1\left(1, \alpha+2\beta; 1+\beta; \frac{1-z}{2-z}\right) \\ &\quad + \rho \frac{4^{1-\alpha-\beta} \Gamma(2\alpha+2\beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma^2(\beta)} z^{2\alpha-1}(1-z)^{2\beta-1} F_2\left(2\alpha+2\beta, 1, 1; \alpha+1, \beta+1; \frac{z}{2}, \frac{1-z}{2}\right) \\ &= f_Z(z; \theta). \end{aligned}$$

Therefore, we have proven that

$$f_{\frac{x}{x+y}}(z) = f_Z(z; \theta), \quad 0 < z < 1. \quad (18)$$

In other words, the stochastic representation for Z given in (6) is valid.

Figure 4 shows the joint PDF of Morgenstern's bivariate distribution with Gamma margins and contour lines of the joint density, for fixed $\alpha = 1.1$ and $\beta = 1.5$ and varying $\rho \in \{-0.99, 0, 0.75\}$. Note that when $\rho = -0.99$, a possible local maximum appears near the global maximum. This corresponds to the same parameter configuration that produced bimodality in one of the plots in Fig. 3.

3.3 Other characterizations of EB model

Let X and Y be two (absolutely) continuous positive random variables with CDFs F_X and F_Y , respectively. Based on F_X and F_Y and its respective PDFs f_X and f_Y , we define two new densities g_X and g_Y as follows:

$$g_X(x) = 2f_X(x)F_X(x), \quad g_Y(y) = 2f_Y(y)F_Y(y), \quad x, y > 0. \quad (19)$$

Proposition 3.1 *The non-negative functions f_1, f_2, f_3 and f_4 defined as (for $s = 1/z - 1$ and $0 < z < 1$)*

$$\begin{aligned} f_1(z) &\equiv (s+1)^2 \int_0^\infty x f_X(x) f_Y(sx) dx, \\ f_2(z) &\equiv (s+1)^2 \int_0^\infty x g_X(x) f_Y(sx) dx, \\ f_3(z) &\equiv (s+1)^2 \int_0^\infty x f_X(x) g_Y(sx) dx, \\ f_4(z) &\equiv (s+1)^2 \int_0^\infty x g_X(x) g_Y(sx) dx, \end{aligned} \quad (20)$$

are PDFs. In the above, g_X and g_Y are as in (19).

Proof Let p and q be two arbitrary densities with positive support. In what follows, we prove that the function g defined as (for $s = 1/z - 1$ and $0 < z < 1$)

$$g(z) \equiv (s+1)^2 \int_0^\infty x p(x) q(sx) dx$$

is a PDF. Indeed, since

$$\int_0^1 g(z) dz = \int_0^\infty x p(x) \left[\int_0^1 \frac{1}{z^2} q\left(\left(\frac{1}{z} - 1\right)x\right) dz \right] dx,$$

taking the change of variable $y = (1/z - 1)x$, the right-hand integral is written as

$$= \int_0^\infty x p(x) \left[\frac{1}{x} \int_0^\infty q(y) dy \right] dx = 1,$$

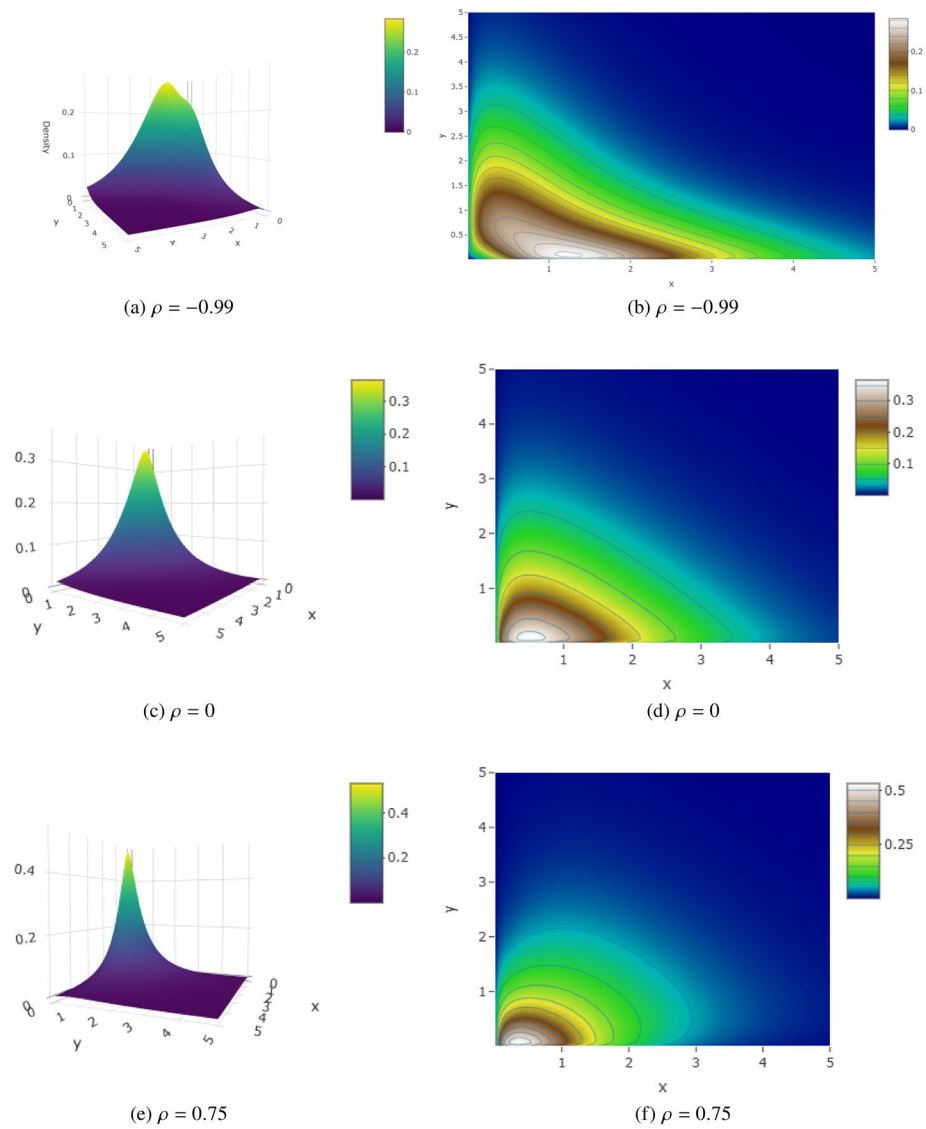


Fig. 4 Joint PDF of Morgenstern's bivariate distribution with Gamma margins and contour lines of the joint density for $\rho = -0.99$ in (a) and (b), $\rho = 0$ in (c) and (d), and $\rho = 0.75$ in (e) and (f). $\alpha = 1.1$ and $\beta = 1.5$

where in the last equality, we have used the fact that p and q are PDFs.

Hence, taking (i) $p = f_X$ and $q = f_Y$, (ii) $p = g_X$ and $q = f_Y$, (iii) $p = f_X$ and $q = g_Y$, and (iv) $p = g_X$ and $q = g_Y$, it follows that f_1, f_2, f_3 and f_4 in (20) are PDFs. $\square$
 Since $f_i(z) = K_i(z)$, $i = 1, 2, 3, 4$, where K_i is as in (11), from (10), we have (for $0 < z < 1$)

$$f_{\frac{X}{X+Y}}(z) = (1 + \rho)f_1(z) - \rho f_2(z) - \rho f_3(z) + \rho f_4(z). \quad (21)$$

If furthermore, $(X, Y)^\top$ follows a Morgenstern's bivariate distribution (with gamma margins), then, by (18) and (21), we have

$$f_Z(z; \theta) = f_{\frac{X}{X+Y}}(z) = \sum_{i=1}^4 \pi_i f_i(z), \quad 0 < z < 1,$$

where $\pi_1 = 1 + \rho$, $\pi_2 = \pi_3 = -\rho$ and $\pi_4 = \rho$. In other words, if Z_i is a continuous positive random variable having CDF $F_{Z_i}(z)$, $0 < z < 1$, $i = 1, 2, 3, 4$, where $X \sim \text{Gamma}(\alpha, \theta)$ and $Y \sim \text{Gamma}(\beta, \theta)$, then Z admits a signed mixture representation; that is, its CDF can be written as the following linear combination:

$$F_Z(z; \theta) = F_{\frac{X}{X+Y}}(z) = \sum_{i=1}^4 \pi_i F_{Z_i}(z), \quad 0 < z < 1. \quad (22)$$

Furthermore, note that the above representation corresponds to a proper convex mixture only when $\rho = 0$.

Proposition 3.2 *Suppose that*

- (a) X_1 and Y_1 are independent variables having densities $f_X(x)$ and $f_Y(x)$, respectively;
- (b) X_2 and Y_2 are independent variables having densities $g_X(x)$ and $f_Y(x)$, respectively;
- (c) X_3 and Y_3 are independent variables having densities $f_X(x)$ and $g_Y(x)$, respectively;
- (d) X_4 and Y_4 are independent variables having densities $g_X(x)$ and $g_Y(x)$, respectively.

If Z_i has PDF $f_i(z)$, $i = 1, 2, 3, 4$, given in (20), then

$$Z_i = \frac{X_i}{X_i + Y_i}, \quad i = 1, 2, 3, 4.$$

Proof The proof follows by using the Jacobian method and then by determining the marginal with respect to Z_i of the random vector $(Z_i, U_i)^\top$, where $Z_i = X_i/(X_i + Y_i)$ and $U_i = Y_i$, $i = 1, 2, 3, 4$. $\square$

By combining (22) and Proposition 3.2, with $(X, Y)^\top$ following a Morgenstern's bivariate distribution (with gamma margins), we get

$$F_Z(z; \theta) = F_{\frac{X}{X+Y}}(z) = \sum_{i=1}^4 \pi_i F_{\frac{X_i}{X_i + Y_i}}(z), \quad 0 < z < 1. \quad (23)$$

3.4 The cumulative distribution function

Using the stochastic representation of $Z \sim \text{EB}(\theta)$, given in (6), by the Law of total expectation we have (for $s = 1/z - 1$ and $0 < z < 1$)

$$\begin{aligned} F_Z(z; \theta) &= \mathbb{P}\left(\frac{X}{X+Y} \leq z\right) = \mathbb{P}(sX \leq Y) = 1 - \mathbb{P}(Y \leq sX) \\ &= 1 - \int_0^\infty \mathbb{P}(Y \leq sX | X = x) f_X(x) dx \\ &= 1 - \int_0^\infty \int_0^{sx} f_{X,Y}(x, y) dy dx. \end{aligned} \quad (24)$$

Since $(X, Y)^\top$ has Morgenstern's bivariate distribution (8) and (9) (with gamma margins) (see Sect. 3.2), the above identities are briefly written as

$$F_Z(z; \theta) = 1 - \int_0^\infty f_X(x) \left\{ F_Y(sx) + \rho[2F_X(x) - 1] \left[\int_0^{sx} 2f_Y(y)F_Y(y)dy - F_Y(sx) \right] \right\} dx. \quad (25)$$

As $Y \sim \text{Gamma}(\beta, \theta)$ and $dF_Y^2(y) = 2f_Y(y)F_Y(y)dy$, we obtain

$$\int_0^{sx} 2f_Y(y)F_Y(y)dy = F_Y^2(sx) = \left[\frac{\gamma(\beta, \theta sx)}{\Gamma(\beta)} \right]^2. \quad (26)$$

As $X \sim \text{Gamma}(\alpha, \theta)$ and $Y \sim \text{Gamma}(\beta, \theta)$, by using (26), $F_Z(z; \theta)$ in (25) is written as

$$\begin{aligned} F_Z(z; \theta) &= 1 - \frac{\theta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} \exp(-\theta x) \left[\frac{\gamma(\beta, \theta sx)}{\Gamma(\beta)} + \rho \left\{ 2 \frac{\gamma(\alpha, \theta x)}{\Gamma(\alpha)} - 1 \right\} \frac{\gamma(\beta, \theta sx)}{\Gamma(\beta)} \left\{ \frac{\gamma(\beta, \theta sx)}{\Gamma(\beta)} - 1 \right\} \right] dx \\ &= 1 - (\rho + 1) \frac{\theta^\alpha}{\Gamma(\alpha)\Gamma(\beta)} \int_0^\infty x^{\alpha-1} \exp(-\theta x) \gamma(\beta, \theta sx) dx \\ &\quad - \rho \frac{2\theta^\alpha}{\Gamma^2(\alpha)\Gamma^2(\beta)} \int_0^\infty x^{\alpha-1} \exp(-\theta x) \gamma(\alpha, \theta x) \gamma^2(\beta, \theta sx) dx \\ &\quad + \rho \frac{2\theta^\alpha}{\Gamma^2(\alpha)\Gamma(\beta)} \int_0^\infty x^{\alpha-1} \exp(-\theta x) \gamma(\alpha, \theta x) \gamma(\beta, \theta sx) dx \\ &\quad + \rho \frac{\theta^\alpha}{\Gamma(\alpha)\Gamma^2(\beta)} \int_0^\infty x^{\alpha-1} \exp(-\theta x) \gamma^2(\beta, \theta sx) dx, \quad s = 1/z - 1. \end{aligned}$$

By using the identities in (13) and (16), we have (for $0 < z < 1$)

$$\begin{aligned} F_Z(z; \theta) &= 1 - (\rho + 1) \frac{\Gamma(\alpha + \beta)}{\beta \Gamma(\alpha)\Gamma(\beta)} (1 - z)^\beta z^\alpha {}_2F_1(\alpha + \beta, 1; \beta + 1; 1 - z) \\ &\quad + \rho \frac{2\Gamma(2\alpha + \beta)}{\alpha\beta \Gamma^2(\alpha)\Gamma(\beta)} \frac{(1 - z)^\beta z^{2\alpha}}{(1 + z)^{2\alpha + \beta}} {}_2F_2\left(2\alpha + \beta, 1, 1; \alpha + 1, \beta + 1; \frac{z}{1 + z}, \frac{1 - z}{1 + z}\right) \\ &\quad + \rho \frac{\Gamma(\alpha + 2\beta)}{\beta^2 \Gamma(\alpha)\Gamma^2(\beta)} \frac{(1 - z)^{2\beta} z^\alpha}{(2 - z)^{\alpha + 2\beta}} {}_2F_2\left(\alpha + 2\beta, 1, 1; \beta + 1, \beta + 1; \frac{1 - z}{2 - z}, \frac{1 - z}{2 - z}\right) \\ &\quad - \rho \frac{2}{\Gamma^2(\alpha)\Gamma^2(\beta)} \int_0^\infty u^{\alpha-1} \exp(-u) \gamma(\alpha, u) \gamma^2\left(\beta, \frac{1 - z}{z} u\right) du. \end{aligned}$$

By using the well-known identity

$$B_x(a, b) = \frac{x^a(1-x)^b}{a} {}_2F_1(1, a+b; 1+a; x),$$

where $B_x(a, b) = \int_0^x t^{a-1}(1-t)^{b-1}dt$ is the incomplete beta function, we write $F_Z(z; \theta)$ as follows (for $0 < z < 1$)

$$\begin{aligned} F_Z(z; \theta) &= 1 - (\rho + 1)I_{1-z}(\beta, \alpha) \\ &+ \rho \frac{2\Gamma(2\alpha + \beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma(\beta)} \frac{(1-z)^\beta z^{2\alpha}}{(1+z)^{2\alpha+\beta}} F_2\left(2\alpha + \beta, 1, 1; \alpha + 1, \beta + 1; \frac{z}{1+z}, \frac{1-z}{1+z}\right) \\ &+ \rho \frac{\Gamma(\alpha + 2\beta)}{\beta^2\Gamma(\alpha)\Gamma^2(\beta)} \frac{(1-z)^{2\beta} z^\alpha}{(2-z)^{\alpha+2\beta}} F_2\left(\alpha + 2\beta, 1, 1; \beta + 1, \beta + 1; \frac{1-z}{2-z}, \frac{1-z}{2-z}\right) \\ &- \rho \frac{2}{\Gamma^2(\alpha)\Gamma^2(\beta)} \int_0^\infty u^{\alpha-1} \exp(-u) \gamma(\alpha, u) \gamma^2\left(\beta, \frac{1-z}{z} u\right) du. \end{aligned} \quad (27)$$

where $I_z(a, b) = B_z(a, b)/B(a, b)$ is the regularized incomplete beta function. To the best of our knowledge, the above integral does not admit a closed-form expression in terms of known special functions; therefore, it must be evaluated numerically.

Remark 3.3 When $\rho = 0$, by using the well-known identity $I_z(\alpha, \beta) = 1 - I_{1-z}(\beta, \alpha)$, we have (for $0 < z < 1$)

$$F_Z(z; \theta) = I_z(\alpha, \beta),$$

which is a very well-known result in the literature.

3.5 Moment-generating function and moments

Given that $Z \sim \text{EB}(\theta)$ has unitary support, its moment-generating function (MGF), $M_Z(t)$, and raw moments, μ_n , $n \in \mathbb{N}$, exist and are finite, particularly within an open interval encompassing zero. Then $M_Z(t)$ may be expanded in a power series about 0 in the following form

$$M_Z(t) = 1 + \sum_{n=1}^{\infty} \frac{\mu_n}{n!} t^n.$$

In what follows, we derive a closed-form expression for μ_n . Indeed, by using the well-known formula for moments of positive order $\mathbb{E}(X^p) = p \int_0^\infty x^{p-1} \mathbb{P}(X > x) dx$ of positive random variables (which simplifies to $\mathbb{E}(X^p) = p \int_0^1 x^{p-1} \mathbb{P}(X > x) dx$ for random variables with support $(0, 1)$, since in this case $\mathbb{P}(X > x) = 0$ for $x \geq 1$), we have

$$\mu_n = n \int_0^1 z^{n-1} [1 - F_Z(z; \theta)] dz,$$

where $F_Z(z; \theta)$ is as in (27). So, from (27),

$$\begin{aligned} \mu_n &= (\rho + 1)n \int_0^1 z^{n-1} I_{1-z}(\beta, \alpha) dz \\ &\quad - \rho \frac{2n\Gamma(2\alpha + \beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma(\beta)} \int_0^1 \frac{(1-z)^\beta z^{2\alpha+n-1}}{(1+z)^{2\alpha+\beta}} F_2\left(2\alpha + \beta, 1, 1; \alpha + 1, \beta + 1; \frac{z}{1+z}, \frac{1-z}{1+z}\right) dz \\ &\quad - \rho \frac{n\Gamma(\alpha + 2\beta)}{\beta^2\Gamma(\alpha)\Gamma^2(\beta)} \int_0^1 \frac{(1-z)^{2\beta} z^{\alpha+n-1}}{(2-z)^{\alpha+2\beta}} F_2\left(\alpha + 2\beta, 1, 1; \beta + 1, \beta + 1; \frac{1-z}{2-z}, \frac{1-z}{2-z}\right) dz \\ &\quad + \rho \frac{2n}{\Gamma^2(\alpha)\Gamma^2(\beta)} \int_0^1 z^{n-1} \int_0^\infty u^{\alpha-1} \exp(-u) \gamma(\alpha, u) \gamma^2\left(\beta, \frac{1-z}{z} u\right) du dz. \end{aligned} \quad (28)$$

For the particular scenario where $\alpha = \beta$, we obtain $\mu_1 = 1/2$ (see Sect. 3.1). Due to the integral's complexity in (28), numerical methods are required for its computation. Since closed-form moments are unavailable, Table 1 provides numerical approximations for μ_n (with $n = 1, \dots, 10, 50, 100$) for different choices of ρ in the special case when $\alpha = \beta = 1$. The numerical computations reported in Table 1 were performed using numerical integration of (25). The results were then compared with those obtained from (28). A more detailed discussion of the numerical implementations will be provided in Sect. 5.7.

4 Estimation

4.1 Maximum likelihood estimation

The log-likelihood function for $\theta = (\alpha, \beta, \rho)^\top$ based on a sample of n independent observations is given by

$$\begin{aligned} \ell(\theta) &= \sum_{i=1}^n \log \left[(1 + \rho) \frac{z_i^{\alpha-1} (1 - z_i)^{\beta-1}}{B(\alpha, \beta)} \right. \\ &\quad - \rho \frac{2\Gamma(2\alpha + \beta)}{\alpha\Gamma^2(\alpha)\Gamma(\beta)} \frac{z_i^{2\alpha-1} (1 - z_i)^{\beta-1}}{(1 + z_i)^{2\alpha+\beta}} {}_2F_1\left(1, 2\alpha + \beta; 1 + \alpha; \frac{z_i}{1 + z_i}\right) \\ &\quad - \rho \frac{2\Gamma(\alpha + 2\beta)}{\beta\Gamma(\alpha)\Gamma^2(\beta)} \frac{z_i^{\alpha-1} (1 - z_i)^{2\beta-1}}{(2 - z_i)^{\alpha+2\beta}} {}_2F_1\left(1, \alpha + 2\beta; 1 + \beta; \frac{1 - z_i}{2 - z_i}\right) \\ &\quad \left. + \rho \frac{4^{1-\alpha-\beta} \Gamma(2\alpha + 2\beta)}{\alpha\beta\Gamma^2(\alpha)\Gamma^2(\beta)} z_i^{2\alpha-1} (1 - z_i)^{2\beta-1} F_2\left(2\alpha + 2\beta, 1, 1; \alpha + 1, \beta + 1; \frac{z_i}{2}, \frac{1 - z_i}{2}\right) \right]. \end{aligned} \quad (29)$$

The maximum likelihood estimator (MLE) $\hat{\theta} = (\hat{\alpha}, \hat{\beta}, \hat{\rho})^\top$ of $\theta = (\alpha, \beta, \rho)^\top$ is obtained by the maximization of the log-likelihood function (29). However, it is not possible to derive an analytical solution for the MLE $\hat{\theta}$; therefore, we must require a numerical solution using some optimization algorithm such as Newton–Raphson and quasi-Newton. But computations of the likelihood $\ell(\theta)$ can be extremely costly due to the dependence of the PDF (2) and CDF (27) on special functions such as the Gauss hypergeometric ${}_2F_1$ and Appell F_2 . Moreover, optimization methods may not converge properly.

Table 1 Moments values for different choices of ρ ($\alpha = \beta = 1$)

ρ	-0.95	-0.9	-0.75	-0.5	0	0.5	0.75	0.9	0.95
μ_1	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500
μ_2	0.348	0.347	0.345	0.341	0.333	0.326	0.322	0.319	0.319
μ_3	0.272	0.271	0.268	0.262	0.250	0.238	0.232	0.229	0.228
μ_4	0.225	0.224	0.220	0.213	0.200	0.187	0.180	0.176	0.175
μ_5	0.193	0.192	0.188	0.181	0.167	0.153	0.146	0.142	0.140
μ_6	0.169	0.168	0.164	0.157	0.143	0.129	0.122	0.118	0.116
μ_7	0.151	0.150	0.146	0.139	0.125	0.111	0.104	0.100	0.099
μ_8	0.137	0.135	0.131	0.124	0.111	0.098	0.091	0.087	0.086
μ_9	0.125	0.123	0.119	0.113	0.100	0.087	0.081	0.077	0.075
μ_{10}	0.115	0.113	0.110	0.103	0.091	0.078	0.072	0.068	0.067
μ_{50}	0.028	0.027	0.026	0.024	0.020	0.015	0.013	0.012	0.011
μ_{100}	0.014	0.014	0.013	0.012	0.010	0.008	0.006	0.006	0.006

Remark 4.1 The Fisher information matrix associated with the parameter vector $\theta = (\alpha, \beta, \rho)^\top$ is defined by

$$I(\theta) = \mathbb{E}[\nabla \log f_Z(Z; \theta) (\nabla \log f_Z(Z; \theta))^\top],$$

where $\nabla = (\partial/\partial\alpha, \partial/\partial\beta, \partial/\partial\rho)^\top$ is the gradient operator and $f_Z(z; \theta)$ is the EB PDF defined in (2).

Observe that, when $\rho \rightarrow 0$,

$$f_Z(z; \theta) \rightarrow f_0(z; \alpha, \beta) = \frac{z^{\alpha-1}(1-z)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 < z < 1,$$

that is, the classical Beta distribution. Consequently, the (α, β) block of the Fisher information matrix coincides with the Fisher information matrix of the Beta distribution, namely,

$$I_0(\alpha, \beta) = \begin{pmatrix} \psi_1(\alpha) - \psi_1(\alpha + \beta) & -\psi_1(\alpha + \beta) \\ -\psi_1(\alpha + \beta) & \psi_1(\beta) - \psi_1(\alpha + \beta) \end{pmatrix},$$

where $\psi_1(\cdot)$ denotes the trigamma function. However, when ρ is treated as an unknown parameter, the full Fisher information matrix remains three-dimensional, since the score function with respect to ρ is generally nonzero even at $\rho = 0$.

The Fisher information matrix for the full parameter vector $\theta = (\alpha, \beta, \rho)^\top$ is therefore

$$I(\theta) = \left(\int_0^1 \frac{\partial \log f_Z(z; \theta)}{\partial \theta_i} \frac{\partial \log f_Z(z; \theta)}{\partial \theta_j} f_Z(z; \theta) dz \right)_{1 \leq i, j \leq 3}, \quad \theta_1 \equiv \alpha, \theta_2 \equiv \beta, \theta_3 \equiv \rho. \quad (30)$$

Since the density $f_Z(z; \theta)$ involves Gauss hypergeometric and Appell functions, the Fisher information matrix $I(\theta)$ does not admit explicit closed-form expressions. Therefore, following Efron and Hinkley (1978), it is preferable to use the observed Fisher information, defined as the negative Hessian matrix, since it is directly computed from the log-likelihood function and typically provides more accurate standard error approximations in finite samples.

4.2 Two-stage plug-in likelihood procedure

Aiming to overcome the computational challenges of the MLE, we provide two alternative estimation approaches. Our first strategy is as follows: given a random sample from $(X, Y)^\top$, we first fit the margins as $\text{Gamma}(\alpha, 1)$ and $\text{Gamma}(\beta, 1)$, and then we estimate ρ in a second step, done using the two-stage plug-in likelihood (PML) estimation. In this case, given an observed sample (x_i, y_i) , $i = 1, \dots, n$, the PML of $\hat{\rho}$ is given by:

$$\hat{\rho} = \arg \max_{\rho} \sum_{i=1}^n \log c(F(x_i), G(y_i); \rho), \quad (31)$$

where $c(\cdot, \cdot; \rho)$ is the copula density and the maximization is taken over the respective parameter space of ρ . Although this procedure does not guarantee the global optimum of the joint MLE, it offers computational advantages in terms of cost and provides a good alternative for obtaining parameter estimates (cf. Lawless and Yilmaz (2011); Lima et al. (2024)).

Thus, comparing X and Y through $Z = X/(X + Y)$, we obtain a natural estimator for the parameters of Z . Naturally, we can first use the estimates from the $\text{Beta}(\alpha, \beta)$ model and then estimate ρ in a second stage, thus improving the initial Beta fit. The Algorithm 2 summarizes it in the two steps. These steps are effective in scenarios where the Beta model provides a good initial fit to the data.

Input: a random sample $\mathbf{Z} = (Z_1, \dots, Z_n)^\top \in [0, 1]^n$.

Output: $\hat{\theta} = (\hat{\alpha}, \hat{\beta}, \hat{\rho})^\top$.

1: Using the $\text{Beta}(\alpha, \beta)$ estimators, compute

$$(\hat{\alpha}, \hat{\beta})^\top = (\hat{\alpha}(\mathbf{Z}), \hat{\beta}(\mathbf{Z}))^\top;$$

2: Estimate ρ using an appropriate optimization algorithm:

$$\hat{\rho} \in \arg \max_{\rho \in [-1, 1]} \ell(\rho), \quad (32)$$

where $\ell(\rho) := \ell(\theta) \mid_{(\alpha, \beta) = (\hat{\alpha}, \hat{\beta})}$;

3: **Return** $\hat{\theta} = (\hat{\alpha}, \hat{\beta}, \hat{\rho})^\top$.

Algorithm 2 Estimation of θ based on improving Beta distribution

When the unit data is obtained from $Z = X/(X + Y)$, then, using Theorem 2 in Lai (1978), we obtain the estimator

$$\hat{\rho} = \frac{\widehat{\text{corr}}(X, Y) \hat{\sigma}_x \hat{\sigma}_y}{\hat{\beta}_x \hat{\beta}_y}, \quad (33)$$

where $\sigma_x^2 = \text{Var } X$, $\sigma_y^2 = \text{Var } Y$, $\beta_x = \int_0^\infty G(x; \alpha, 1) [1 - G(x; \alpha, 1)] dx$, and

$$\beta_y = \int_0^\infty G(x; \beta, 1) [1 - G(x; \beta, 1)] dx,$$

in which $G(\cdot; \theta, 1)$ denotes the CDF of a Gamma model with shape θ and scale 1. It is easy to see that

$$\hat{\rho} = \frac{\pi}{\alpha\beta} \frac{\widehat{\text{cov}}(X, Y) \Gamma(\alpha + 1) \Gamma(\beta + 1)}{\Gamma(\alpha + 1/2) \Gamma(\beta + 1/2)}.$$

Consequently, Algorithm 3 can be used. If necessary, (X, Y) are rescaled to ensure that they share the same scale parameter.

Input: a random sample $(\mathbf{X}, \mathbf{Y}) = ((X_1, Y_1)^\top, \dots, (X_n, Y_n)^\top)^\top \in ((0, \infty) \times (0, \infty))^n$.

Output: $\hat{\theta} = (\hat{\alpha}, \hat{\beta}, \hat{\rho})^\top$.

1: Using the $\text{Gamma}(\alpha, \theta)$ estimators, compute the shape parameters

$$(\hat{\alpha}, \hat{\beta})^\top = (\hat{\alpha}(\mathbf{X}), \hat{\beta}(\mathbf{Y}))^\top;$$

2: Estimate ρ using (31) or (33);

3: **Return** $\hat{\theta} = (\hat{\alpha}, \hat{\beta}, \hat{\rho})^\top$.

Algorithm 3 Estimation of θ based on improving Beta distribution

Remark 4.2 The second step (31) in Algorithm 3 requires optimization procedures. A natural initial guess is (33). Using the Spearman correlation $\rho_S = \frac{\rho}{3} \in [-1/3, 1/3]$,

we can choose the starting values $\rho_0 \approx 3\hat{\rho}_S$. Similarly, we could use the Kendall estimate ($\tau = 2/9\rho \in [-2/9, 2/9]$).

5 Simulation study

5.1 Random sample generator

From (8), the joint distribution function $F_{X,Y}(x, y)$ is obtained by composing the copula $C(\cdot, \cdot)$, defined in (7), with the marginal Gamma CDFs $F_X(x)$ and $F_Y(y)$, where $X \sim \text{Gamma}(\alpha, \theta)$ and $Y \sim \text{Gamma}(\beta, \theta)$. That is,

$$F_{X,Y}(x, y) = C(F_X(x), F_Y(y)), \quad x, y > 0.$$

Aiming to simulate a random sample $Z_1, \dots, Z_n$ from the extended Beta PDF (2), we use the Algorithm 4, which simulates $Z = X/(X + Y) \sim f_Z(\cdot; \theta)$ based on a simulation of $(X, Y) \sim F_{X,Y}$. For a further discussion on this algorithm, we refer the reader to Mai and Scherer (2017).

Input: $\theta = (\alpha, \beta, \rho)^\top \in (0, \infty)^2 \times [-1, 1]$.

Output: $(X, Y) \sim F_{X,Y}$ and $Z \sim f_Z(\cdot; \theta)$.

1: Simulate $U_2 \sim U[0, 1]$.

2: Compute the partial derivative

$$F_{U_1|U_2}(u_1) := \left. \frac{\partial}{\partial u_2} C(u_1, u_2) \right|_{u_2=U_2} = u_1 [1 + \rho(1 - u_1)(1 - 2u_2)], \quad u_1 \in [0, 1].$$

3: Compute the generalized inverse $F_{U_1|U_2}^{-1}$ numerically:

$$F_{U_1|U_2}^{-1}(v) := \inf \{u_1 > 0; F_{U_1|U_2}(u_1) \geq v\}, \quad v \in (0, 1),$$

4: Get a sample $V \sim U[0, 1]$, independent of U_2 .

5: Define $U_1 := F_{U_1|U_2}^{-1}(V)$ and take (U_1, U_2) as the random vector of the copula C .

6: Compute $(X, Y) \sim F_{X,Y}$ as

$$(X, Y)^\top = (F_X^{-1}(U_1), F_Y^{-1}(U_2))^\top,$$

where $F_{X,Y}$ is given in (8) and F_X and F_Y denote the CDFs of Gamma models as described in Subsection 3.4.

7: **Return** (X, Y) and $Z = \frac{X}{X+Y}$.

Algorithm 4 Computational sampling of the extended Beta distribution

5.2 R implementation and required packages

For this study, all implementations were performed in R software (R Core Team 2025).

We implement the Appell function F_2 in R using the double series (3). For other function implementations, a wider range may be required. In such cases, numerical integral representations can be employed. Some additional formulas for the Appell function F_2 can be found in Brychkov and Saad (2014) and Ananthanarayan et al. (2023). The generalized hypergeometric function is computed using `hyperg_2F1` from the `gsl` package. Thus, the PDF (2) can be implemented in R.

Referring to Algorithm 4, the numerical inverse is required. We achieve this using the function `uniroot`.

A gradient-based method can be applied to optimize the log-likelihood function. The gradient of the objective function is obtained using the function `grad` in the package `numDeriv`.

Remark 5.1 Another simulation algorithm could be proposed by directly inverting the CDF (27) and applying it to a standard uniform random variable. Comments on CDF implementation are given in 5.7.

5.3 Simulation results for the two-stage PML estimator

To evaluate the performance of the two-stage PML estimator $\hat{\theta}$, we fix several values of the parameters $\theta = (\alpha, \beta, \rho)^\top$, and then generate $N = 1,000$ Monte-Carlo samples with sample size $n \in \{30, 100, 300, 1000\}$, of the random variable $Z \sim \text{EB}(\alpha, \beta, \rho)$. We analyze the PML estimates $\hat{\theta}$ (Algorithm 3) through relative bias (RB), and root mean squared error (RMSE).

We use the gradient descent algorithm to achieve the minimum of the minus log-likelihood function (cf. Shalev-Shwartz and Ben-David 2014) to obtain the PML estimates for

the EB distribution. The gradient of the objective function can be obtained numerically, as described in Sect. 5.2.

For the simulation, the following procedure was carried out:

-
- Input:** $\theta = (\alpha, \beta, \rho)^\top \in (0, \infty)^2 \times [-1, 1]$, n (sample size), N (number of Monte Carlo simulations)
Output: RB and RMSE of ML estimates.
- 1: **for** $j = 1$ to N **do**
 - 2: Simulate a Monte-Carlo sample $(\mathbf{X}, \mathbf{Y})^{(j)} = ((X_1, Y_1)^{(j)}, \dots, (X_n, Y_n)^{(j)})$ via Algorithm 4.
 - 3: Compute $\hat{\theta}^{(j)} = \hat{\theta}^{(j)}((\mathbf{X}, \mathbf{Y})^{(j)})$, using Algorithm 3.
 - 4: **end for**
 - 5: Compute the RB and RMSE as follows:
-

$$RB(\hat{\theta}_k) = \frac{1}{N} \sum_{j=1}^N \frac{\hat{\theta}_k^{(j)} - \theta_k}{\theta_k}, \quad k = 1, 2, 3, \text{ and } \theta_1 = \alpha, \theta_2 = \beta, \theta_3 = \rho,$$

and

$$RMSE(\hat{\theta}_k) = \sqrt{\frac{1}{N} \sum_{j=1}^N (\hat{\theta}_k^{(j)} - \theta_k)^2}.$$

- 6: **Return** $RB(\hat{\theta}_k)$ and $RMSE(\hat{\theta}_k)$ for $k = 1, 2, 3$, and $\theta_1 = \alpha, \theta_2 = \beta, \theta_3 = \rho$.
-

Algorithm 5 Assessing PML estimates via Monte Carlo simulation

The simulation results are presented in Figs. 5 and 6. The PML estimator shows good behavior with minimal RB and low RMSE. Furthermore, it is clear that increasing the sample size n leads RB and RMSE to approach zero. This demonstrates the precision of the PML estimator for the UF distribution, even with small sample sizes. Additionally, we observe that the RMSE of ρ estimates is less affected by variations in sample size. In general, for $n \geq 100$, the RB is below 0.10 and converges to zero as n increases. Although relatively low (in absolute values), the relative bias of ρ exhibited a different sign compared to the estimates of the parameters α and β .

Regarding the RMSE, large values were observed in some cases for small sample sizes. However, the value decreased when larger samples were taken ($n \in \{300, 1000\}$). The RMSE of the β estimates was high (≥ 0.5) when $n = 30$, in both cases $\rho = -0.5$ and $\rho = 0.5$. When $n = 100$, the values decreased to 0.20, stabilizing below 0.10 only for $n \geq 300$.

In Fig. 7, we fixed $\theta^\top = (2, 3, 0.5)$ and plotted the histogram of Monte Carlo simulation results of estimates. When $n = 30$ (top), the estimates of the three parameters were right-skewed. We can see that as n increases, the figures become symmetric, showing the good asymptotic behavior of the estimates. The same behavior occurs with the other parameters studied.

Although gradient descent is a simple optimization algorithm, it may not be the most suitable choice for bounded-parameter likelihoods. Figure 8 compares the estimation of ρ using gradient descent and the L-BFGS-B method. The two approaches exhibit similar behavior, except in the extreme cases when $\rho = \pm 0.95$, when the L-BFGS-B performs better for large sample sizes.

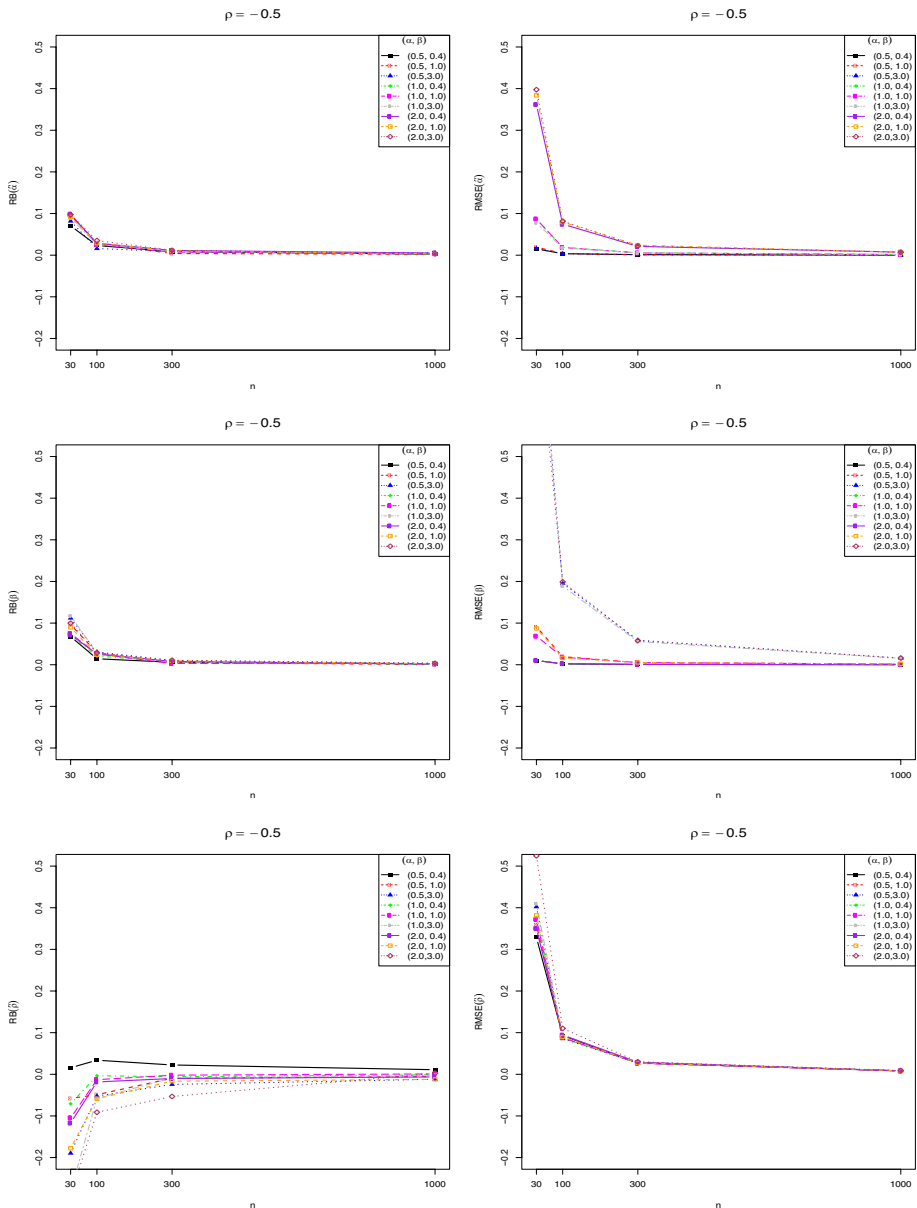


Fig. 5 RB (left) and RMSE (right) for the PML estimates of α (top), β (middle), and ρ (bottom), with varying (α, β) and fixing negative $\rho = -0.5$

5.4 Identifiability through Fisher information

In the previous subsection, some parameter configurations, such as $\theta = (2, 3, 0.5)^\top$, were associated with relatively large RMSE values for small sample sizes. Motivated by this behavior, we investigate the presence of weak non-identifiability in a neighborhood of θ .

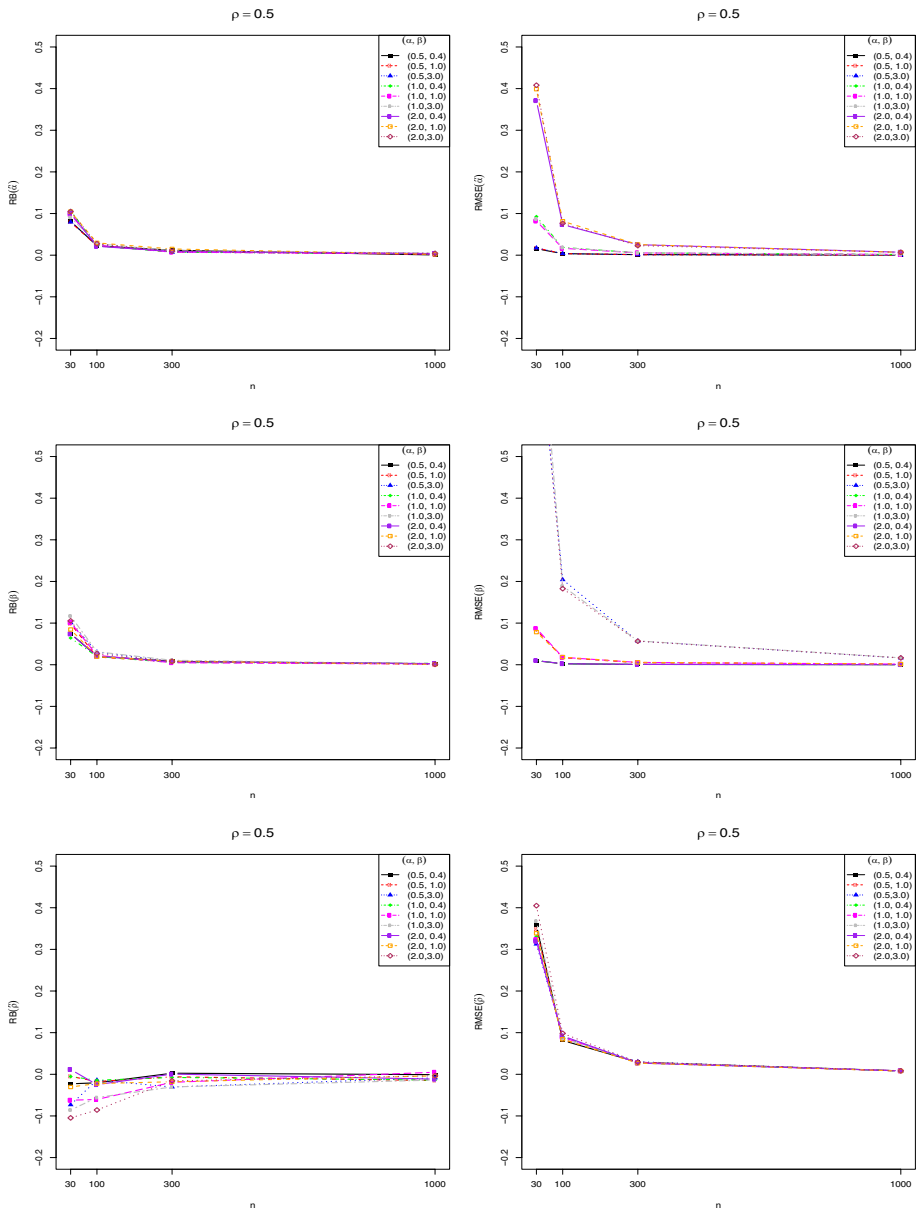


Fig. 6 RB (left) and RMSE (right) for the PML estimates of α (top), β (middle), and ρ (bottom), with varying (α, β) and fixing positive $\rho = 0.5$

To this end, we numerically evaluate the Fisher information matrix (30) for $\theta_0 = (2, 3, \rho)^\top$, varying $\rho \in (-1, 1)$. Specifically, we analyze the following indicators of weak non-identifiability: the proximity of the smallest eigenvalue to zero ($\lambda_{\min} \approx 0$); the condition number ($\kappa(I(\theta_0)) = \lambda_{\max}/\lambda_{\min} \uparrow \infty$); and the proximity of the determinant values to zero ($\det(I(\theta_0)) \approx 0$). Tables 2 and 3 present the results for $|\rho| \approx 1$ and $|\rho| \approx 0$, respectively.

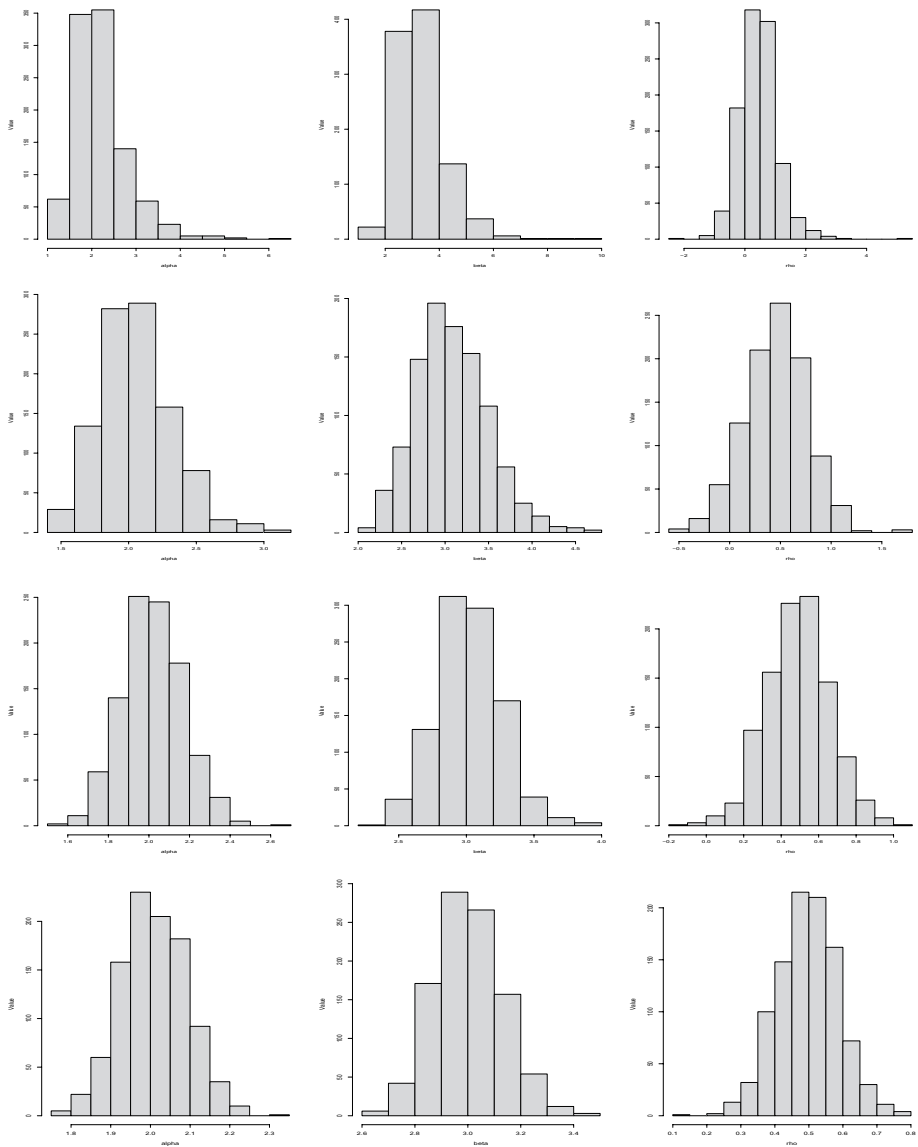


Fig. 7 Histograms of Monte Carlo results for the PML estimates of α (left), β (middle), and ρ (right), with increasing sample size $n = 30$ (top), 100, 300, and 1000 (bottom)

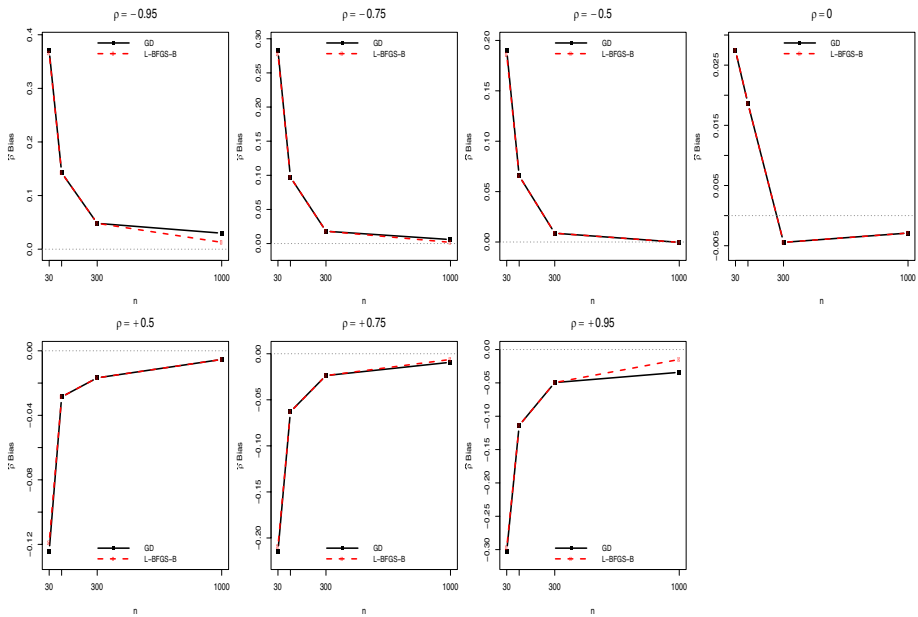


Fig. 8 Monte Carlo results for the Bias decay of PML estimates of $\rho \in \{\pm 0.95, \pm 0.75, \pm 0.5, 0\}$, with increasing sample size $n = 30, 100, 300, 1000$

As $|\rho| \uparrow 1$, $\lambda_{\min}$ appears to remain essentially unchanged with respect to ρ , while $\kappa(I(\theta_0))$ and $\det(I(\theta_0))$ remain above the thresholds of 10^{-6} and below 10^6 , respectively. Therefore, these measures do not indicate non-identifiability. However, these measures exhibit different scales for $\rho \in \{-0.75, 0, 0.75\}$. Then, a more detailed investigation was conducted in the neighborhood of $|\rho| \approx 0$, which likewise revealed no indication of weak non-identifiability.

5.5 Simulation results for the MLE

As highlighted in the previous subsections, the parameter region $\theta = (2, 3, \rho)$ with $\rho \in (-1, 1)$ poses a considerable challenge, exhibiting high RMSE for small sample sizes. This configuration is therefore adopted for the subsequent analysis.

We now evaluate the MLE obtained by maximizing the log-likelihood function (29). The optimization was performed using the “L-BFGS-B” method implemented in the optim function. For this purpose, we consider the following performance measures: Bias, RMSE, coverage probability (CP), computational time in seconds (CT), and the percentage of non-convergence (PNC).

Table 4 reports the MLE results for $\rho \in \{0, \pm 0.75, \pm 0.5\}$, based on $M = 1,000$ Monte Carlo replications with sample sizes $n \in \{30, 50, 100\}$. Despite the high computational cost, the bias and RMSE remain relatively large, even for $n = 100$, particularly for the estimates of ρ . The performance deteriorates further for large positive values of $\rho \in \{0.75, 0.95\}$. The PNC is low, indicating that the algorithms converge reliably, although the sample sizes are insufficient to ensure accurate estimation. The CP values are consistently high, suggesting that the Hessian-based confidence intervals are overly wide.

Table 2 Determinant, condition number, and minimum eigenvalue of $I(\theta_0)$ for $\theta_0 = (2, 3, \rho)$, varying $\rho \in \{\pm 0.99, \pm 0.95, \pm 0.90, \pm 0.75, 0\}$

ρ	$\det(I(\theta_0))$	$\kappa(I(\theta_0))$	$\lambda_{\min}$
-0.99	3.083×10^{-4}	51.16	0.00849
-0.95	3.006×10^{-4}	53.03	0.00824
-0.90	2.911×10^{-4}	55.54	0.00793
-0.75	2.640×10^{-4}	64.44	0.00701
0.00	1.398×10^{-4}	184.68	0.00295
+0.75	4.066×10^{-5}	1261.85	0.00056
+0.90	4.958×10^{-5}	1242.86	0.00060
+0.95	6.078×10^{-5}	1083.94	0.00071
+0.99	7.461×10^{-5}	934.03	0.00083

Table 3 Determinant and condition number of $I(\theta_0)$ for $\theta_0 = (2, 3, \rho)$, varying $\rho \in \{\pm 0.10, \pm 0.05, \pm 0.01, 0\}$

ρ	$\det(I(\theta_0))$	$\kappa(I(\theta_0))$
-0.10	1.560×10^{-4}	154.19
-0.05	1.479×10^{-4}	168.40
-0.01	1.414×10^{-4}	181.24
0	1.398×10^{-4}	184.68
+0.01	1.382×10^{-4}	188.22
+0.05	1.317×10^{-4}	203.44
+0.10	1.236×10^{-4}	225.18

Table 5 presents the sensitivity of the MLE to different starting values. For the true parameter $\theta = (2, 3, 0)^\top$, $M = 1000$ Monte Carlo replications were conducted using five fixed initializations: the true value $(2, 3, 0)^\top$, a value close to the true parameter $(2.5, 3.5, 0.2)^\top$, a standard parameter $(1, 1, 0)^\top$, a distant parameter $(0.5, 0.5, 0.8)^\top$, and an opposite direction $(5, 5, -0.8)^\top$. The Bias and RMSE were not influenced by the choice of initial value.

5.6 Comparison of estimators

Consider a random sample $(X, Y) = ((X_1, Y_1)^\top, \dots, (X_n, Y_n)^\top)^\top$, where (X, Y) follows a Morgenstern's bivariate distribution with gamma margins (8). Then, we can generate the sample $Z = (Z_1, \dots, Z_n) \sim \text{EB}(\theta)$ from the ratio (6). In this subsection, we compare the MLE results in Table 4 with the two-stage PMLE, using Algorithm 3.

Table 6 shows that the PMLE (Algorithm 3) exhibits behavior similar to the MLE in the estimation of α and β . However, the PMLE was more accurate in the estimation of ρ , while requiring considerably less computational time. On the other hand, the coverage probability for ρ performs worse when compared with the MLE results. However, this “worse” behavior implies that the standard errors (SEs) are larger for ML estimates than for PML estimates. For the CP computations, the standard errors (SEs) were obtained as the square root of the inverse of the Hessian matrix. For the MLE, the Hessian was computed using the optim function at the optimization point. For the FMLE, the SEs of the first stage were derived from the fitdist output, while in the second stage $\text{SE}(\hat{\rho}(\hat{\alpha}, \hat{\beta}))$ was calculated from the square root of the inverse of the resulting Hessian. In real data applications, to mitigate

Table 4 Monte Carlo results for the MLE of $\theta = (2, 3, \rho)$, $\rho \in \{0, \pm 0.75, \pm 0.95\}$, with increasing sample size $n \in \{30, 50, 100\}$

Parameter	n	Bias	RMSE	CP	CT	PNC
$\alpha = 2$	30	0.05600	0.58956	0.9905	767.20	0.00
	50	-0.02656	0.43703	0.9857	1411.89	0.20
	100	-0.06042	0.32979	0.9843	2351.14	0.30
$\beta = 3$	30	0.08018	0.88569	0.9893	767.20	0.00
	50	-0.05671	0.69583	0.9879	1411.89	0.20
	100	-0.10326	0.52213	0.9823	2351.14	0.30
$\rho = -0.95$	30	0.33787	0.70863	0.9964	767.20	0.00
	50	0.35338	0.71369	0.9890	1411.89	0.20
	100	0.29689	0.62501	0.9864	2351.14	0.30
$\alpha = 2$	30	0.11124	0.63604	0.9917	748.53	0.00
	50	0.02210	0.46759	0.9890	1255.88	0.20
	100	-0.02313	0.35448	0.9864	2426.31	0.20
$\beta = 3$	30	0.16851	0.95772	0.9917	748.53	0.00
	50	0.02206	0.74557	0.9868	1255.88	0.20
	100	-0.04233	0.56078	0.9832	2426.31	0.20
$\rho = -0.75$	30	0.21429	0.70100	0.9964	748.53	0.00
	50	0.23314	0.70115	0.9901	1255.88	0.20
	100	0.19733	0.63401	0.9843	2426.31	0.20
$\alpha = 2$	30	0.35182	0.79184	0.9964	854.45	0.20
	50	0.25137	0.64850	0.9946	1414.72	0.30
	100	0.12106	0.47688	0.9903	2775.91	0.40
$\beta = 3$	30	0.56009	1.22529	0.9976	854.45	0.20
	50	0.39668	1.03399	0.9946	1414.72	0.30
	100	0.19406	0.75783	0.9924	2775.91	0.40
$\rho = 0$	30	-0.24408	0.82535	1.0000	854.45	0.20
	50	-0.20709	0.78711	1.0000	1414.72	0.30
	100	-0.10590	0.72040	0.9968	2775.91	0.40
$\alpha = 2$	30	0.51064	0.87196	1.0000	866.50	0.10
	50	0.43773	0.75617	1.0000	1459.34	0.20
	100	0.32610	0.61528	0.9814	2966.69	0.20
$\beta = 3$	30	0.81821	1.36724	1.0000	866.50	0.10
	50	0.69083	1.17904	0.9986	1459.34	0.20
	100	0.52790	0.98317	0.9787	2966.69	0.20
$\rho = 0.75$	30	-0.46352	0.92941	1.0000	866.50	0.10
	50	-0.46818	0.90648	1.0000	1459.34	0.20
	100	-0.40425	0.81429	0.9322	2966.69	0.20
$\alpha = 2$	30	0.58795	0.94845	1.0000	895.58	0.10
	50	0.47393	0.78774	0.9984	1470.57	0.30
	100	0.36103	0.62723	0.9829	2868.99	0.50
$\beta = 3$	30	0.94834	1.50294	1.0000	895.58	0.10
	50	0.75136	1.23516	0.9935	1470.57	0.30
	100	0.58462	1.00159	0.9720	2868.99	0.50
$\rho = 0.95$	30	-0.53292	0.92458	1.0000	895.58	0.10
	50	-0.48614	0.85692	0.9967	1470.57	0.30
	100	-0.42980	0.76656	0.9068	2868.99	0.50

Table 5 Monte Carlo results for the MLE of $\theta = (2, 3, 0)$, with fixed sample size $n = 100$, and varying the initial values

Initial values	Parameter	Bias	RMSE	CP	PNC
$(2, 3, 0)^\top$	α	0.12106	0.47688	0.9903	0.40
	β	0.19406	0.75783	0.9924	0.40
	ρ	-0.10590	0.72040	0.9968	0.40
$(2.5, 3.5, 0.2)^\top$	α	0.12386	0.48056	0.9892	0.30
	β	0.19880	0.76614	0.9914	0.30
	ρ	-0.10771	0.71892	0.9968	0.30
$(1, 1, 0)^\top$	α	0.11381	0.47975	0.9902	0.40
	β	0.18071	0.76330	0.9924	0.40
	ρ	-0.09876	0.72179	0.9967	0.40
$(0.5, 0.5, 0.8)^\top$	α	-0.09965	0.43539	0.9884	0.40
	β	-0.17015	0.68812	0.9913	0.40
	ρ	0.29312	0.77951	0.9956	0.40
$(5, 5, -0.8)^\top$	α	0.13310	0.50888	0.9882	0.30
	β	0.20552	0.77643	0.9903	0.30
	ρ	-0.11266	0.72013	0.9968	0.30

the impact of first-step errors on the second-step SEs, a parametric Bootstrap can be applied. In Sect. 6, a real-data application is presented for a large sample size. Due to the computational cost, the PMLE emerges as the preferable approach.

5.7 Computational issues

We conclude this section by highlighting some computational challenges related to the implementation and validation of both the model and its estimators.

- Series truncation and evaluation accuracy: The double series (3), which defines the Appell F_2 function for $|x| + |y| < 1$, can be computationally demanding. For our purposes, we restrict the double summation to a maximum of 200 terms per index and adopt a tolerance parameter of $\varepsilon = 10^{-12}$. To ensure accuracy, we compared selected configurations with the `mpmath.appellf2` function in Python, obtaining differences below 10^{-6} in all tests. In addition, some well-known identities were verified, yielding satisfactory results.
- Numerical issues encountered in practice: The PDF and CDF of the EB model can be computationally intensive when applied to large datasets or in likelihood-based estimation procedures. Pre-computed lookup tables for special functions can provide an efficient alternative for practical implementations.
- Extreme parameter values: When $|\rho| \uparrow 1$, the estimation yields poorer results. The model may become inapplicable when the data (X, Y) are strongly associated, leading to the generation of a ratio random variable Z .
- Numerical Computation of the CDF: Although the CDF $F_Z(z; \theta)$ was not used in simulations, it will be applied in real data analyses. For straightforward numerical computation, we implemented (25) using the `integrate` function in R with specifications `rel.tol = 1e - 8, abs.tol = 1e - 12, subdivisions = 1000L`, supported by the `pgamma` function. To assess the accuracy of the closed-form expression (27), we compared its results with those obtained from (25) and from direct numerical integration of the PDF (2). Table 11 in Appendix C reports the numerical results across a grid of parameter

Table 6 Monte Carlo results for the PMLE of $\theta = (2, 3, \rho)$, $\rho \in \{0, \pm 0.75, \pm 0.95\}$, with increasing sample size $n \in \{30, 50, 100\}$

Parameter	n	Bias	RMSE	CP	CT	PNC
$\alpha = 2$	30	0.20498	0.61157	0.9569	1.34	0.20
	50	0.12671	0.42604	0.9639	1.80	0.30
	100	0.06193	0.27979	0.9567	3.13	0.60
$\beta = 3$	30	0.29971	0.94568	0.9619	1.34	0.20
	50	0.16644	0.64704	0.9619	1.80	0.30
	100	0.07672	0.43218	0.9437	3.13	0.60
$\rho = -0.95$	30	0.36592	0.64695	0.4749	1.34	0.20
	50	0.24496	0.45351	0.5416	1.80	0.30
	100	0.14261	0.29062	0.5775	3.13	0.60
$\alpha = 2$	30	0.20647	0.61553	0.9529	1.39	0.20
	50	0.12889	0.43260	0.9599	1.92	0.30
	100	0.06171	0.28041	0.9567	3.48	0.60
$\beta = 3$	30	0.29971	0.94568	0.9619	1.39	0.20
	50	0.16644	0.64704	0.9619	1.92	0.30
	100	0.07672	0.43218	0.9437	3.48	0.60
$\rho = -0.75$	30	0.27773	0.61676	0.6202	1.39	0.20
	50	0.17840	0.45499	0.6901	1.92	0.30
	100	0.09727	0.30915	0.8048	3.48	0.60
$\alpha = 2$	30	0.21488	0.63130	0.9549	1.39	0.30
	50	0.13538	0.45079	0.9548	2.27	0.40
	100	0.06143	0.28454	0.9538	3.67	0.50
$\beta = 3$	30	0.30062	0.94596	0.9619	1.39	0.30
	50	0.16670	0.64736	0.9618	2.27	0.40
	100	0.07704	0.43204	0.9437	3.67	0.50
$\rho = 0$	30	0.02746	0.55469	0.8195	1.39	0.30
	50	0.02814	0.44205	0.8886	2.27	0.40
	100	0.01863	0.31960	0.9156	3.67	0.50
$\alpha = 2$	30	0.22374	0.64224	0.9599	1.50	0.20
	50	0.13919	0.45725	0.9569	2.08	0.30
	100	0.06213	0.28982	0.9467	3.75	0.50
$\beta = 3$	30	0.30077	0.94560	0.9619	1.50	0.20
	50	0.16744	0.64766	0.9619	2.08	0.30
	100	0.07691	0.43177	0.9437	3.75	0.50
$\rho = 0.75$	30	-0.20944	0.50820	0.6293	1.50	0.20
	50	-0.12734	0.37565	0.7141	2.08	0.30
	100	-0.06316	0.25994	0.8040	3.75	0.50
$\alpha = 2$	30	0.22562	0.64469	0.9580	1.70	0.10
	50	0.13936	0.45787	0.9559	2.14	0.20
	100	0.06227	0.29133	0.9448	3.96	0.30
$\beta = 3$	30	0.30171	0.94593	0.9620	1.70	0.10
	50	0.16743	0.64736	0.9619	2.14	0.20
	100	0.07771	0.43243	0.9438	3.96	0.30
$\rho = 0.95$	30	-0.29735	0.51256	0.5035	1.70	0.10
	50	-0.19327	0.36092	0.5561	2.14	0.20
	100	-0.11379	0.22801	0.5807	3.96	0.30

values. The differences among the three implementations are negligible (less than 10^{-4}), allowing us to use any of them. For the computations in Table 1 and in the following section, the numerical implementation of (25) was used.

These computational challenges and limitations do not compromise the consistent results established for the PMLE and the EB model.

6 Real-world data analysis

In this section, we present an application of the proposed model to real data for illustrative purposes. To ensure replicability of our results and provide support for future users of the EB model, all R codes are publicly available at https://github.com/FSQuintino/extended_beta_model.git (accessed on 07 Jan 2026). A comparison with the well-known Beta and Kumaraswamy models is presented. We also consider a broad range of models with one, two, three, and four parameters, including bimodal distributions that have become well established in the recent literature. For the reader's convenience, in Appendix B we present the PDFs of the competing models.

6.1 Income-consumption data

The dataset comes from the Bank of Italy's Survey on Household Income and Wealth from the year 2008. We utilized data from the RFAM08 data set (with income as the Y variable) and the RISFAM08 data set (with expenditures as the C variable), as outlined in the Survey on Household Income and Wealth 2008 data description file. We excluded any data entries where income or consumption was negative or unavailable. The data were recently reported in Vila and Quintino (2026) and the references therein, and are available at https://www.bancaditalia.it/statistiche/tematiche/indagini-famiglie-imprese/bilanci-famiglie/documentazioni/ricerca/ricerca.html?min_anno_publicazione=2008&max_anno_publicazione=2008 (accessed on 20 November 2025).

In this study, we analyze income-consumption data by constructing a new variable $Z = Y/(X + Y)$ based on (1), where X and Y represent expenditures and income. Values of $Z < 1/2$ indicate that the family ended the year with more expenditures than income. If $Z > 1/2$, the opposite is true. The case $Z = 1/2$ means expenditures and income are equal.

We are interested in modeling Z because it is a one-dimensional bounded variable, easy to interpret, and it summarizes the income-consumption profile of families in a single value. The expected behavior is a unit distribution with left skewness. A full characterization of Z allows us to describe families more effectively and offers an alternative way to study copula-based stress-strength probabilities $\mathbb{P}(X < Y)$, avoiding the limitations of Morgenstern's copula in cases of weak dependence. We then show that the EB model fits Z appropriately.

6.2 Evaluating the EB for the ratio data

In Table 7, we present the following descriptive statistics: the sample size (n); the minimum value observed in the dataset (Min.); the first quartile value (1st Qu.); the median value (Median); the average value (Mean); the third quartile value (3rd Qu.); the maximum value

Table 7 Descriptive statistics for the income-consumption dataset

n	Min	1st Qu	Median	Mean	3rd Qu	Max	Sd	CS	CK
7957	0.00	0.51	0.55	0.56	0.61	0.94	0.09	-0.42	3.34

Table 8 Parameter estimates and model selection (AIC and BIC) for the income-consumption dataset

Model	$\hat{\theta}^\top$	$\ell(\hat{\theta})$	AIC	BIC
EB	(16.10, 12.84, 0.78)	7922.92	-15839.84(*)	-15818.90(*)
Topp-Leone	4.26	7204.62	-14407.25	-14400.27
Unit-Lindley	1.09	3620.73	-7239.47	-7232.49
Beta	(16.10, 12.84)	7832.86	-15661.72	-15647.76
Kumaraswamy	(6.46, 28.86)	7795.81	-15587.63	-15573.66
Logit-Normal	(0.37, 0.87)	4340.08	-8676.16	-8662.19
Unit-Gompertz	(1.81, 0.75)	2069.28	-4134.56	-4120.60
Unit-Inverse Gaussian	(1.37, 5.10)	12011.25	-24018.49	-24004.53
Unit-Weibull	(2.63, 2.48)	5737.09	-11470.19	-11456.22
Unit-Fréchet	(1.25, 5.52, 0.1949)	8381.26	-16756.52	-16735.57
UBBS	(7.15, 0.76, 18.60)	4673.84	-9341.68	-9311.76
Bimodal Beta	(14.86, 13.27, 15.23, 17.46)	7874.68	-15741.36	-15713.44

(*) Since the EB estimates were obtained by PML, the AIC and BIC values were computed by evaluating the log-likelihood (29) at $\hat{\theta}$. Therefore, these criteria are conservative selection measures

observed (Max.); the standard deviation (Sd.); the coefficient of symmetry (CS); the coefficient of kurtosis (CK).

To estimate the EB parameters for dataset Z, we applied (33), after fitting a beta distribution. Consider the ratio variable

$$Z = \frac{Y}{Y + X}.$$

Table 8 gives the estimates, the Akaike information criterion (AIC), and the Bayesian information criterion (BIC) for the fitted models. Since the AIC and BIC values are smaller for the EB model than for the traditional Beta and Kumaraswamy models, this new model seems to be a competitive fit for these data. Among the recently proposed unit distributions reported in Table 8, only the Unit-Inverse Gaussian (UIG) and Unit-Fréchet (UF) distributions exhibit information criterion values lower than those of the EB model. The Bimodal Beta (BBeta) distribution also provides a competitive fit, with information criterion values close to those obtained for the EB model. However, the histogram and empirical CDF (ECDF) plots, together with the fitted PDFs and CDFs (respectively, Figs. 9 and 10), indicate that the UIG model adheres less closely to the empirical distribution than the EB, UF, and BBeta models. This conclusion is supported by the distance measures between the ECDF and the theoretical CDF in Table 9, the ECDF differences in Fig. 11, and the probability integral transform (PIT) in Fig. 12.

Remark 6.1 The parameter estimates of the EB model reported in Table 8 were obtained using the two-stage PML approach (after fitting the beta distribution, (33) was applied to obtain the estimate of ρ), requiring only 0.8467 min of computation time. Although MLE

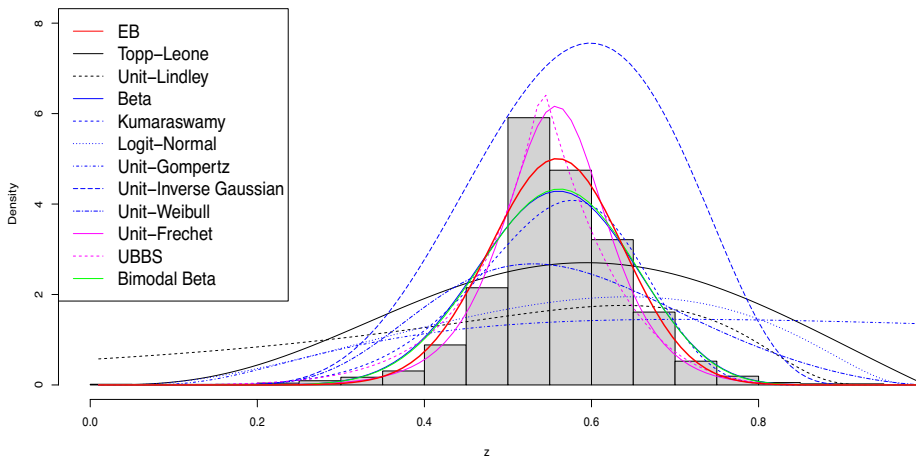


Fig. 9 Histogram with the EB density (red) and competing one-parameter (black), two-parameter (blue), three-parameter (magenta), and four-parameter (green) models fitted to the income-consumption dataset

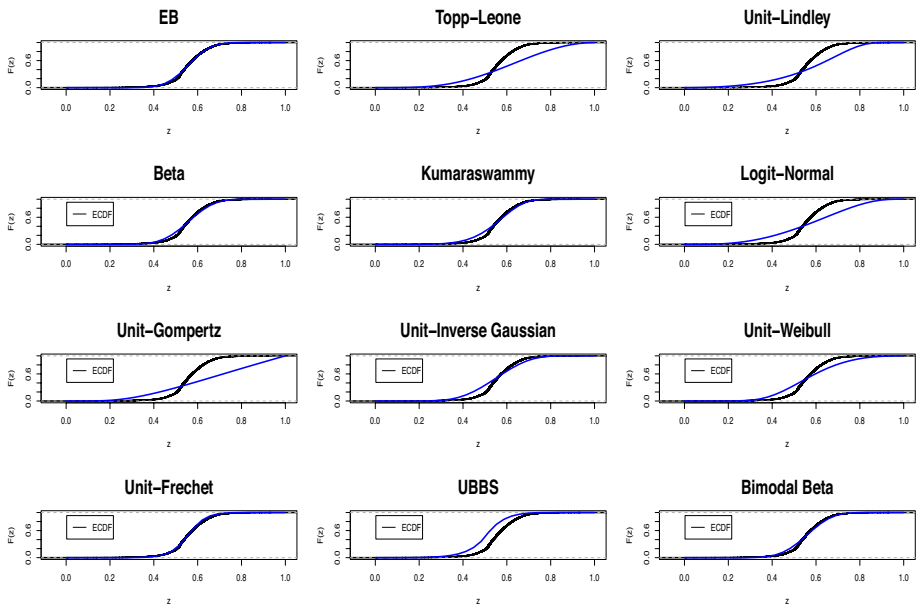
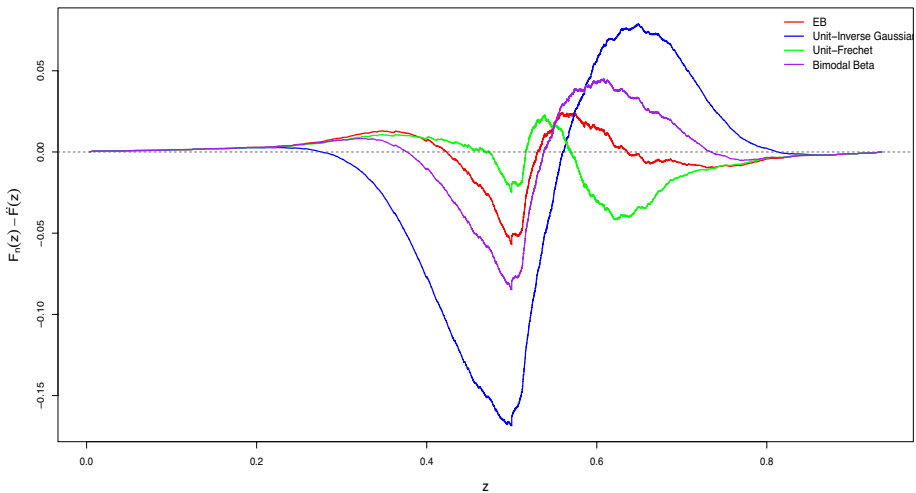


Fig. 10 Empirical CDF (ECDF) compared with EB and competing models for the income-consumption dataset

can also be employed, its computational cost becomes substantial for large datasets. In the present application, MLE required 413.8503 min to report results. While MLE increased the log-likelihood from $\ell(\hat{\theta}^{PMLE}) = 7922.9$ to $\ell(\hat{\theta}^{MLE}) = 195973.8$, the “L-BFGS-B” algorithm failed to converge, returning parameter estimates located on the boundary of the search space.

Table 9 Distance measures: the integrated L_1 distance, the integrated L_2 distance, the supremum distance, and the distance between the PIT and the Unif distribution

Model	L_1	L_2	Sup	PIT vs Unif
EB	0.02	0.02	0.06	0.06
Unit-Inverse Gaussian	0.07	0.09	0.17	0.17
Unit-Fréchet	0.02	0.02	0.04	0.04
Bimodal Beta	0.03	0.04	0.08	0.08

**Fig. 11** Difference between the ECDF and the fitted CDF, $F_n(z) - \hat{F}(z; \hat{\theta})$, for the EB (red), Unit-Inverse Gaussian (blue), Unit-Fréchet (green), and Bimodal Beta (purple) models. Curves closer to the horizontal gray line at zero indicate a better fit

To ensure a good model fit, we use randomized quantile residuals, defined by $R_i = \Phi^{-1}(F(z_i; \hat{\theta}))$, where $\hat{\theta}$ is the estimated parameter, $F(z_i; \hat{\theta})$ is the CDF of the model fitted to each observation z_i , and Φ^{-1} represents the quantile of the standard normal distribution $N(0, 1)$. Comparing the previously selected models (EB, UF, and EB), Figs. 13 and 14 show the randomized quantiles are quite similar, with the UF model providing a slightly better fit in the left tail.

Given the large sample size ($n = 7957$), formal goodness-of-fit tests are expected to be highly sensitive. They may strongly reject the null hypothesis even for models that provide an adequate fit in practical terms. Therefore, greater emphasis was placed on graphical diagnostics and information criteria when evaluating model performance, which showed that the EB model can be an interesting model for that data.

To test the null hypothesis $H_0 : \rho = 0$ (H_0 : beta model) against the alternative hypothesis $H_1 : \rho \neq 0$ (H_1 : EB model), we use the LR statistic, the value of which is 180.12. The two-step approach does not allow us to rely on the known distribution of the LR statistic. Therefore, we employed a parametric bootstrap to compute the p value. After $B = 1,000$ replications, we obtain a p value < 0.01 . Thus, using any usual significance level, we reject the null hypothesis in favor of the EB model, i.e., this model is significantly better than the beta model for explaining the current data.

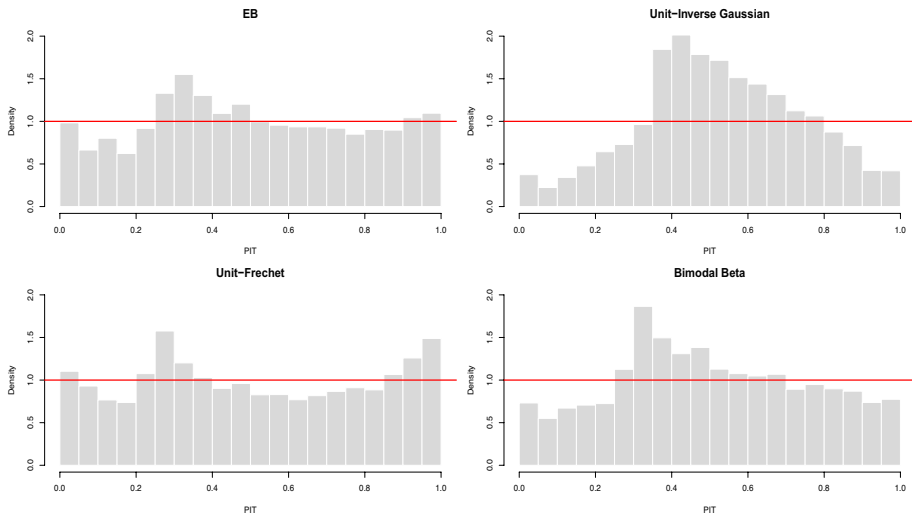


Fig. 12 Probability integral transform (PIT) histograms for the EB (top left), Unit-Inverse Gaussian (top right), Unit-Fréchet (bottom left), and Bimodal Beta (bottom right) models. Histograms closer to the reference uniform density (horizontal red line at 1) indicate a better fit since, under a correctly specified model, the PIT values are uniformly distributed on (0, 1)

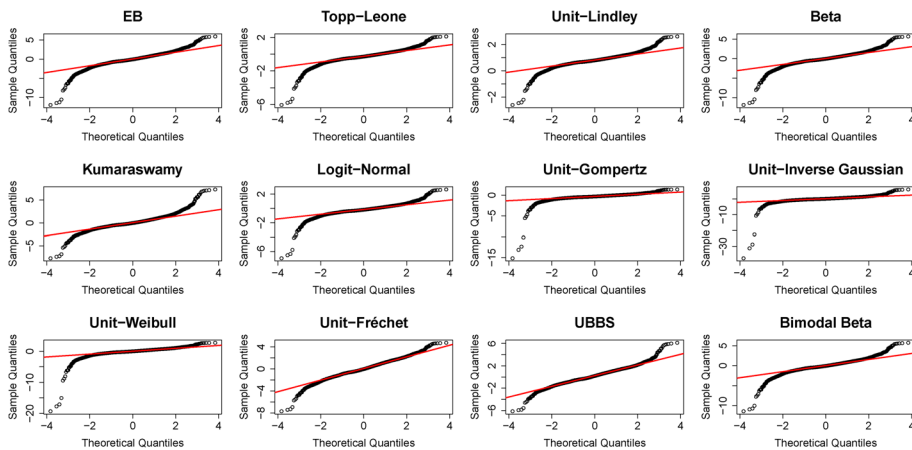


Fig. 13 Quantile-Quantile plot displaying randomized quantile residuals from fitted models for income-consumption data

6.3 Assessing copula selection in income–consumption modeling

As pointed out in Sect. 3.2, the EB model arises from a ratio, where the random vector $(X, Y)^\top$ follows the Morgenstern bivariate distribution with gamma margins. Figure 15 shows the gamma fits of the marginal distributions of expenditures and income. These findings offer more support for the EB model's good fit to the derived Z unitary data, which was discussed in detail in the previous subsection. To estimate the Gamma parameters of

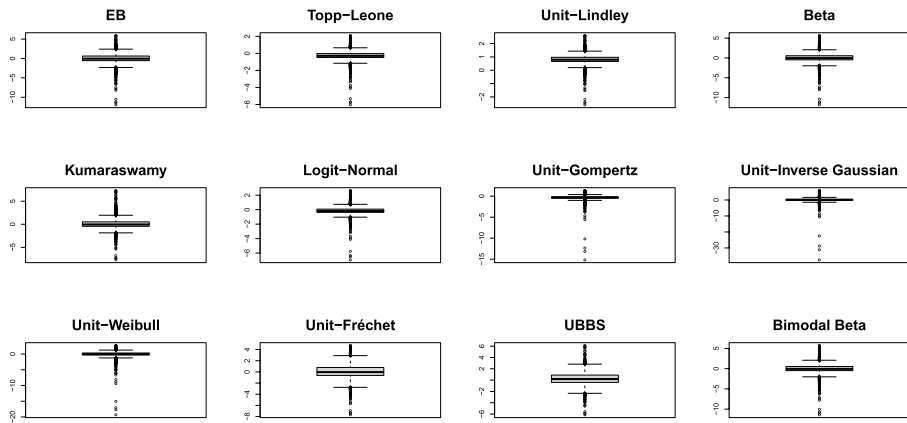


Fig. 14 Boxplot displaying randomized quantile residuals from fitted models for income-consumption data

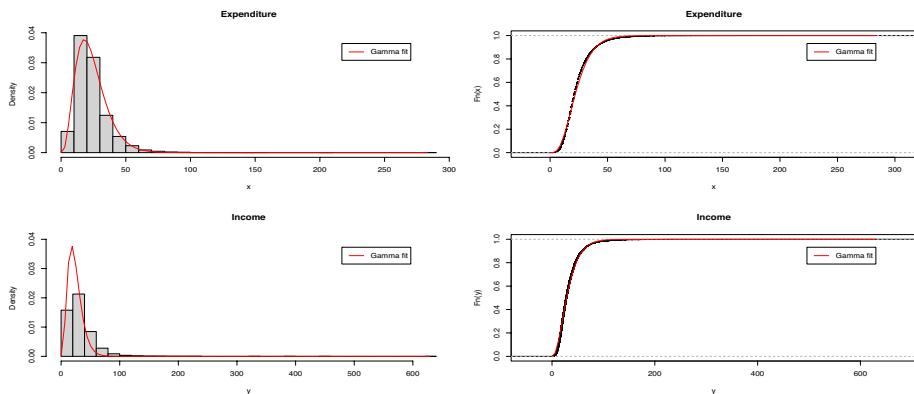


Fig. 15 Gamma fits to marginal distributions of expenditures and income

the margins, we use the `fitdistr` function from the R package `MASS`. Due to the scale of the sample, the estimation procedure exhibits numerical instabilities. Then, the adjustment $(X \cdot 10^{-3}, Y \cdot 10^{-3})$ is required to achieve numerical convergence in the estimation procedures. The choice of 10^3 is based on trials with 10^k . By varying k , the estimation for X becomes numerically tractable at $k = 1$, whereas Y requires $k = 2$. However, the estimated parameters remain too close to zero, leading us to choose $k = 3$.

After fitting the margins and aiming to use the estimator (33) of ρ , we require that the data (X, Y) share the same scale parameter θ . We then transformed the data as follows:

$$x \leftarrow 0.5\hat{\alpha}x/1000 \text{ and } y \leftarrow 0.5\hat{\beta}y/1000.$$

Figure 16 presents simulated samples overlaid on the observed data, along with the corresponding density contours. The model captures the joint behaviour of the data well, although a few outliers can be observed.

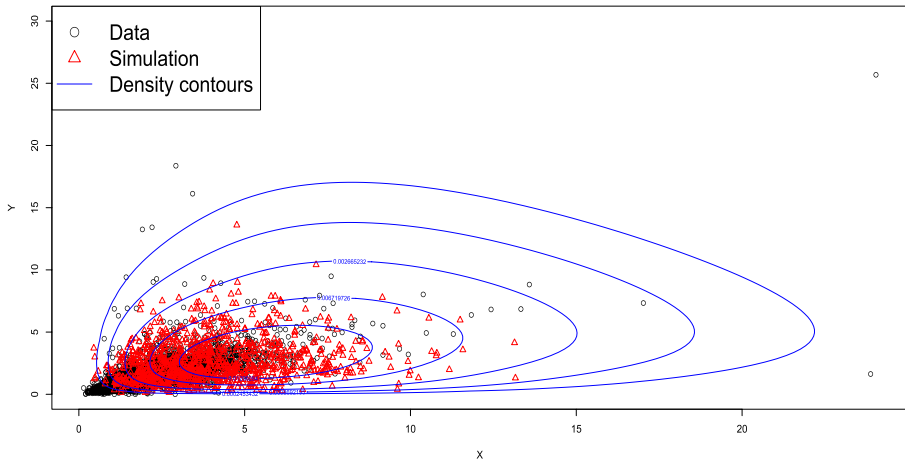


Fig. 16 Simulated samples of $C(\text{Gamma}(4.05, 2), \text{Gamma}(2.65, 2); 0.77)$ (red) overlaid on the observed income–consumption data (black), along with the corresponding density contours (blue lines)

Table 10 Copula dependence estimates, model selection based on the maximum log-likelihood value, and Spearman (ρ_S) and Kendall (τ) corresponding estimates

GFGM generator	$\hat{\theta}$	$\ell(\hat{\theta})$	$\rho_S(\hat{\rho})$	$\tau(\hat{\rho})$
$\phi(x) = \min(x, 1 - x)$	(4.05, 2.65, 0.59)	−51830.61	0.44	0.29
$\phi(x) = x(1 - x)$	(4.05, 2.65, 0.77)	−51532.43	0.26	0.17
$\phi(x) = x(1 - x)(1 - 2x)$	(4.05, 2.65, 1.00)	−52361.27	0(**)	0(**)
$\phi(x) = \frac{1}{\pi} \sin(\pi x)$	(4.05, 2.65, 1.00)	−51054.64	0.49	0.33

(**) The value is zero regardless of the estimate of ρ

Table 10 presents a comparative study involving some more flexible copulas from the extended FGM literature. For the reader’s convenience, in Appendix B we present the CDFs of the benchmark copulas, named the Generalized Farlie-Gumbel-Morgenstern (GFGM) copula. A more detailed revision of GFGM copula, the reader can find information in Amblard and Girard (2002); Susam (2022). In Table 10, the log-likelihood values reveal no substantial differences among the GFGM generators. However, two of them yielded $\hat{\rho} = 1$, suggesting that the model may not be suitable for capturing the dependence structure. Alternative copula families will be considered in future work. Here, we consider the correlation estimates

$$\rho_S(\hat{\rho}) = 12\hat{\rho} \left(\int_0^1 \phi(x) dx \right)^2 \quad \text{and} \quad \tau(\hat{\rho}) = 8\hat{\rho} \left(\int_0^1 \phi(x) dx \right)^2.$$

7 Concluding remarks

In this paper, we introduced the EB distribution, a new unit-supported model obtained from the ratio of two correlated Gamma random variables following a Morgenstern-type dependence structure. The additional dependence parameter extends the classical Beta distribution while preserving interpretability. As shown throughout the paper, the EB family exhibits considerable flexibility in terms of distributional shape, encompassing symmetric and asymmetric densities, as well as boundary-inflated and other nonstandard behaviors that may arise in practice. Closed-form expressions were obtained for the probability density and cumulative distribution functions and moments, relying on tools from special-function theory. Estimation was addressed using likelihood-based and alternative methods, and a Monte Carlo study highlighted both the finite-sample behavior of the estimators and the computational challenges that may arise. Applications to real datasets demonstrated that the EB distribution can provide improved fits compared to the traditional Beta and Kumaraswamy benchmarks, despite requiring only one additional parameter. Despite its flexibility, the proposed model has some limitations. The dependence structure is restricted to the Farlie–Gumbel–Morgenstern copula, which was adopted because of its analytical tractability and the resulting closed-form expressions. Consequently, stronger dependence structures cannot be accommodated within the present framework. In addition, the evaluation of some distributional quantities relies on special functions, which may increase the computational cost of likelihood-based inference. Future research includes regression extensions for responses in $(0, 1)$, Bayesian inference with informative priors, multivariate generalizations, and the development of unit models based on copulas capable of capturing stronger dependence, such as the Gumbel–Hougaard, Frank, and Clayton copulas with Gamma margins. Faster numerical strategies for likelihood maximization also deserve further investigation. Overall, the EB model provides a flexible and analytically tractable addition to the family of unit distributions, offering a useful alternative for modeling bounded data in applications such as economics, reliability, hydrology, and related fields.

Appendix A: Some additional results

The purpose of this section is to present and show some new technical results used throughout the work.

Proposition A.1 *The following identity is valid*

$$\begin{aligned} & \int_0^\infty x^{a-1} \exp(-sx) \gamma(b, \theta x) \gamma(c, \xi x) dx \\ &= \frac{\theta^b \xi^c \Gamma(a+b+c)}{bc(s+\theta+\xi)^{a+b+c}} F_2 \left(a+b+c, 1, 1; b+1, c+1; \frac{\theta}{s+\theta+\xi}, \frac{\xi}{s+\theta+\xi} \right), \end{aligned}$$

where $a > 0$, $s > 0$, $\theta > 0$, $\xi > 0$, $b > 0$ and $c > 0$.

Proof By using the definition of lower incomplete gamma function, we have

$$\gamma(b, \theta x) = \int_0^{\theta x} y^{s-1} \exp(-y) dy = \theta^b x^b \int_0^1 w^{b-1} \exp(-\theta x w) dw,$$

where the change of variable $w = y/(\theta x)$ was considered. Analogously,

$$\gamma(c, \xi x) = \xi^c x^c \int_0^1 u^{c-1} \exp(-\xi x u) du.$$

Hence,

$$\begin{aligned} & \int_0^\infty x^{a-1} \exp(-sx) \gamma(b, \theta x) \gamma(c, \xi x) dx \\ &= \theta^b \xi^c \int_0^\infty \int_0^1 \int_0^1 [x(s + \theta w + \xi u)]^{a+b+c-1} \\ & \quad \times \exp[-x(s + \theta w + \xi u)] \frac{w^{b-1} u^{c-1}}{(s + \theta w + \xi u)^{a+b+c-1}} dw du dx. \end{aligned}$$

Making the change of variable $z = x(s + \theta w + \xi u)$ and changing the order of integration, the last integral becomes

$$\begin{aligned} &= \theta^b \xi^c \int_0^1 \int_0^1 \left[\int_0^\infty z^{a+b+c-1} \exp(-z) dz \right] \frac{w^{b-1} u^{c-1}}{(s + \theta w + \xi u)^{a+b+c}} dw du \\ &= \theta^b \xi^c \Gamma(a + b + c) \int_0^1 \int_0^1 \frac{w^{b-1} u^{c-1}}{(s + \theta w + \xi u)^{a+b+c}} dw du. \end{aligned}$$

By changing w to $1 - w$ and u to $1 - u$, the above integral can be written as

$$= \frac{\theta^b \xi^c \Gamma(a + b + c)}{(s + \theta + \xi)^{a+b+c}} \int_0^1 \int_0^1 \frac{(1-w)^{b-1} (1-u)^{c-1}}{\left(1 - \frac{\theta}{s+\theta+\xi} w - \frac{\xi}{s+\theta+\xi} u\right)^{a+b+c}} dw du.$$

In short, we have proven that

$$\begin{aligned} & \int_0^\infty x^{a-1} \exp(-sx) \gamma(b, \theta x) \gamma(c, \xi x) dx \\ &= \frac{\theta^b \xi^c \Gamma(a + b + c)}{(s + \theta + \xi)^{a+b+c}} \\ & \quad \int_0^1 \int_0^1 \frac{(1-w)^{b-1} (1-u)^{c-1}}{\left(1 - \frac{\theta}{s+\theta+\xi} w - \frac{\xi}{s+\theta+\xi} u\right)^{a+b+c}} dw du. \end{aligned} \quad (34)$$

By using in (34) the well-known integral representation of the Appell F2 double hypergeometric function:

$$\begin{aligned}
 & F_2(a, b_1, b_2; c_1, c_2; x, y) \\
 &= \frac{\Gamma(c_1)\Gamma(c_2)}{\Gamma(b_1)\Gamma(b_2)\Gamma(c_1-b_1)\Gamma(c_2-b_2)} \\
 & \int_0^1 \int_0^1 u^{b_1-1} v^{b_2-1} (1-u)^{c_1-b_1-1} (1-v)^{c_2-b_2-1} (1-ux-vy)^{-a} du dv,
 \end{aligned}$$

the proof follows. $\square$

Proposition A.2 *The following identity is valid*

$$\int_0^\infty x^{a-1} \exp(-sx) \gamma(b, \theta x) dx = \frac{\theta^b \Gamma(a+b)}{b(s+\theta)^{a+b}} {}_2F_1\left(a+b, 1; b+1; \frac{\theta}{s+\theta}\right),$$

where $a > 0$, $s > 0$, $\theta > 0$ and $b > 0$.

Proof By taking $c = 1$ and then $\xi \rightarrow \infty$ in (34), we get

$$\begin{aligned}
 & \int_0^\infty x^{a-1} \exp(-sx) \gamma(b, \theta x) dx = \lim_{\xi \rightarrow \infty} \frac{\theta^b \xi \Gamma(a+b+1)}{(s+\theta+\xi)^{a+b+1}} \\
 & \int_0^1 \int_0^1 \frac{(1-w)^{b-1}}{\left(1 - \frac{\theta}{s+\theta+\xi} w - \frac{\xi}{s+\theta+\xi} u\right)^{a+b+1}} dw du \\
 &= \lim_{\xi \rightarrow \infty} \theta^b \Gamma(a+b) \int_0^1 (1-w)^{b-1} \left\{ (s+\theta-\theta w)^{-a-b} - (s+\theta+\xi-\theta w)^{-a-b} \right\} dw \\
 &= \frac{\theta^b \Gamma(a+b)}{(s+\theta)^{a+b}} \int_0^1 (1-w)^{b-1} \left(1 - \frac{\theta}{s+\theta} w\right)^{-a-b} dw.
 \end{aligned} \tag{35}$$

By using in (35) the well-known integral representation of the hypergeometric function:

$${}_B(b, c-b) {}_2F_1(a, b; c; z) = \int_0^1 x^{b-1} (1-x)^{c-b-1} (1-zx)^{-a} dx, \quad c > b > 0,$$

the proof readily follows. $\square$

Appendix B: Benchmark models

For the reader's convenience, we present below the CDFs of the competitors' copulas and the PDFs of the unit competing models.

B.1 Bivariate copula models

The CDF of the Generalized Farlie-Gumbel-Morgenstern copula is given by:

$$C^{GFGM}(u, v; \rho) = uv + \rho \phi(u) \phi(v), \quad \rho \in [-1, 1],$$

where u and v are the marginal distribution functions of random variables X and Y , respectively. Observe that if $\rho = 0$, then $C^{FGM}(u, v; 0) = uv$, that is, X and Y are independent.

To get the PDF associated with each CDF and use it in the estimation procedures, it suffices to consider that

$$c(u, v; \rho) = \frac{\partial^2 C(u, v; \rho)}{\partial u \partial v}.$$

B.2 Unit models

For simplicity of notation, we set the parameters as α , β , and ρ for all competing distributions. The ML estimates for the models below were obtained using the `goodness.fit` function from the `AdequacyModel` package in R software.

The Beta and Kumaraswamy models have PDFs, respectively, given by

$$f_B(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1, \alpha > 0, \beta > 0,$$

and

$$f_K(x; \alpha, \beta) = \alpha\beta x^{\alpha-1} (1-x^\alpha)^{\beta-1}, \quad 0 < x < 1, \alpha > 0, \beta > 0.$$

The one-parameter competing models used in Sect. 6 are the Unit-Lindley and the Topp-Leone, which have PDFs given, respectively, by:

$$\begin{aligned} f_{UL}(x; \alpha) &= \frac{\alpha^2}{1+\alpha} (1-x)^{-3} \exp\left(-\frac{\alpha x}{1-x}\right), \quad 0 < x < 1, \alpha > 0 \\ f_{TL}(x) &= 2\alpha x^{\alpha-1} (1-x)(2-x)^{\alpha-1}, \quad 0 < x < 1, \alpha > 0. \end{aligned}$$

The other two-parameter competing models used in Sect. 6 are the Logit-Normal, Unit-Gompertz, Unit-inverse Gaussian, and Unit-Weibull, which have PDFs given, respectively, by:

$$\begin{aligned} f_{LN}(x; \alpha, \beta) &= \frac{1}{\beta\sqrt{2\pi}} \exp\left[-\frac{(\text{logit}(x) - \alpha)^2}{2\beta^2}\right] \frac{1}{x(1-x)}, \quad 0 < x < 1, \alpha \in \mathbb{R}, \beta > 0, \\ f_{UG}(x; \alpha, \beta) &= \alpha\beta x^{-(\beta+1)} \exp[-\alpha(x^{-\beta} - 1)], \quad 0 < x < 1, \alpha > 0, \beta > 0, \\ f_{UIG}(x; \alpha, \beta) &= \sqrt{\frac{\beta}{2\pi}} (x(1-x))^{-3/2} \exp\left(-\frac{\beta(x - \alpha(1-x))^2}{2\alpha^2 x(1-x)}\right), \quad 0 < x < 1, \alpha > 0, \beta > 0, \\ f_{UW}(x; \alpha, \beta) &= \frac{1}{x} \alpha\beta (-\log x)^{\beta-1} \exp[-\alpha(-\log x)^\beta], \quad 0 < x < 1, \alpha > 0, \beta > 0. \end{aligned}$$

The three-parameter competing models used in Sect. 6 are the Unit-Fréchet and Unit-Bimodal Birnbaum-Saunders, which have PDFs given, respectively, by:

$$f_{UF}(x; \theta) = \frac{\alpha}{\beta^\alpha} s^{\alpha-1} (s+1)^2 \left\{ \frac{2 \left(\frac{s^\alpha}{\beta^\alpha} + 1 \right)^2 - \rho \left[\left(\frac{s^\alpha}{\beta^\alpha} \right)^2 + 1 \right]}{\left[\left(\frac{s^\alpha}{\beta^\alpha} + 1 \right)^2 - \rho \frac{s^\alpha}{\beta^\alpha} \right]^2} - \frac{1}{\left(\frac{s^\alpha}{\beta^\alpha} + 1 \right)^2} \right\}, \quad s = \frac{x}{1-x},$$

where $\alpha > 0, \beta > 0$, and $0 \leq \rho \leq 1$, and

$$f_{UBBS}(x; \alpha, \beta, \rho) = \frac{1}{4x\alpha\beta\Phi(x)} \left[\left(-\frac{\beta}{\log x} \right)^{1/2} + \left(-\frac{\beta}{\log x} \right)^{3/2} \right] \phi(|t(x)| + \rho),$$

where $x \in (0, 1)$, $t(x) = \frac{1}{\alpha} \left(\sqrt{-\frac{\log x}{\beta}} - \sqrt{-\frac{\beta}{\log x}} \right)$, $\alpha > 0, \beta > 0, \rho \in \mathbb{R}$.

The four-parameter competing model used in Sect. 6 is the Bimodal Beta distribution, which has PDFs given by:

$$f_{BB}(x; \alpha, \beta, \rho, \delta) = \frac{\rho + (1 - \delta x)^2}{Z B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1,$$

where $\alpha > 0, \beta > 0, \rho \geq 0, \delta \in \mathbb{R}$,

$$Z = Z(\alpha, \beta, \rho, \delta) = 1 + \rho - 2\delta \frac{\alpha}{\alpha + \beta} + \delta^2 \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)}$$

denotes the normalization constant and $B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt$ is the beta function.

Appendix C

In this section, we present Table 11 to access the numerical computation of the CDF (27). Three numerical versions were computed: the closed form (27), the alternative CDF via numerical integration (25) and the numerical integration of the PDF (2). Some parameters θ are chosen as well as the z values. The codes for replications are available on the GitHub page reported in Sect. 6.

Table 11 Numerical evaluation of the CDF across different implementations

$\theta^\top$	z	Closed-form (27)	Alternative CDF (25)	PDF integration
(2, 3, 0.7)	0.4	0,5284683460	0,5284683726	0,5284683726
(1.5, 0.8, -0.6)	0.65	0,4475317167	0,4475317166	0,4475317166
(3.2, 2.1, 1)	0.25	0,0200278508	0,0201862658	0,0201862658
(0.7, 1.3, -1)	0.8	0,8790286820	0,8790286820	0,8790286820
(2.5, 2.5, 0.35)	0.5	0,4999999987	0,5000000000	0,5000000000
(1, 1, 0.9)	0.3	0,2657890395	0,2657918552	0,2657918552

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest There are no conflict of interest to disclose.

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